

Compal Confidential

KALH0 /KAL90+ /KALG0 M/B Schematics Document

Intel Penryn Processor with Cantiga + DDRIII + ICH9M

2009-3-4

REV:1.0

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Model Name : KALH0/KALG0/KAL90+

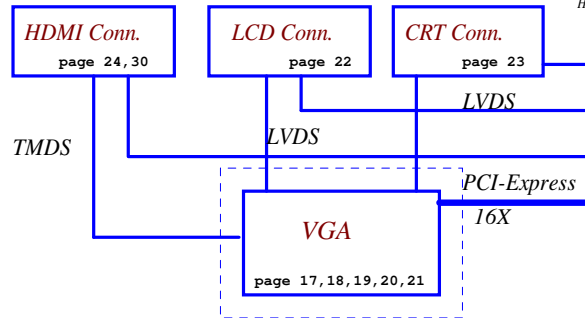
Fan Control
page 40

Intel Penryn Processor
uPGA-478 Package
(Socket P) page 4, 5, 6

Thermal Sensor
EMC 1402
page 4

Clock Generator
ICS9LPRS387
page 16

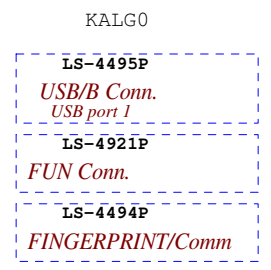
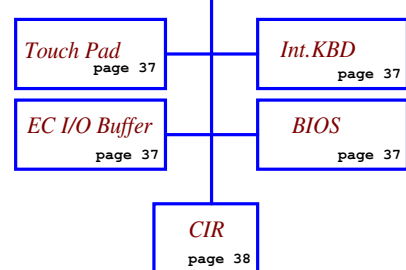
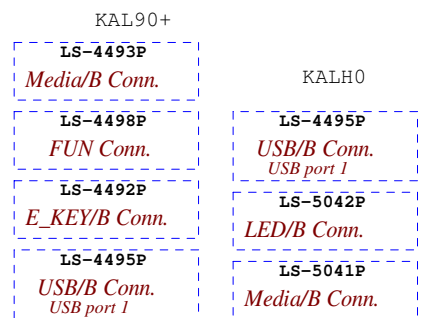
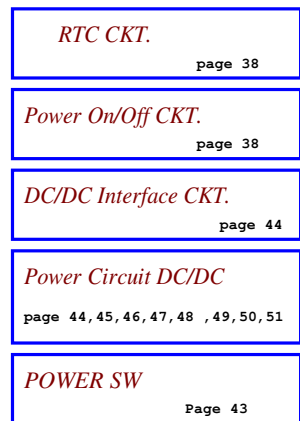
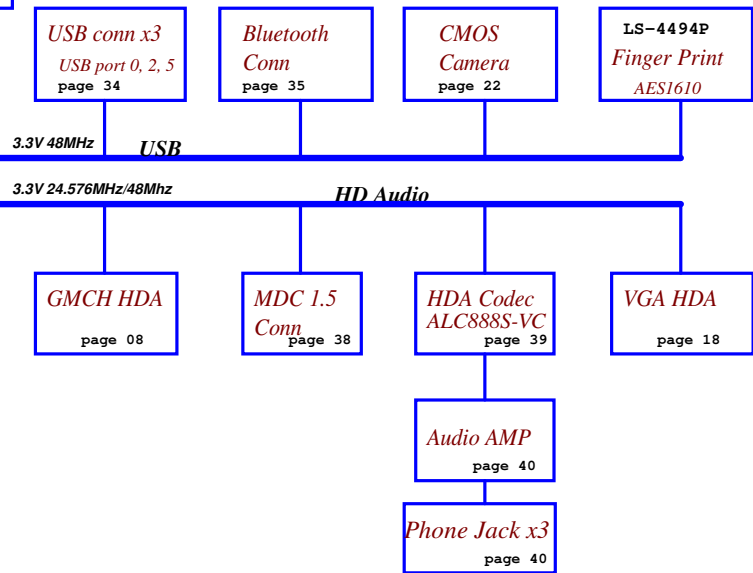
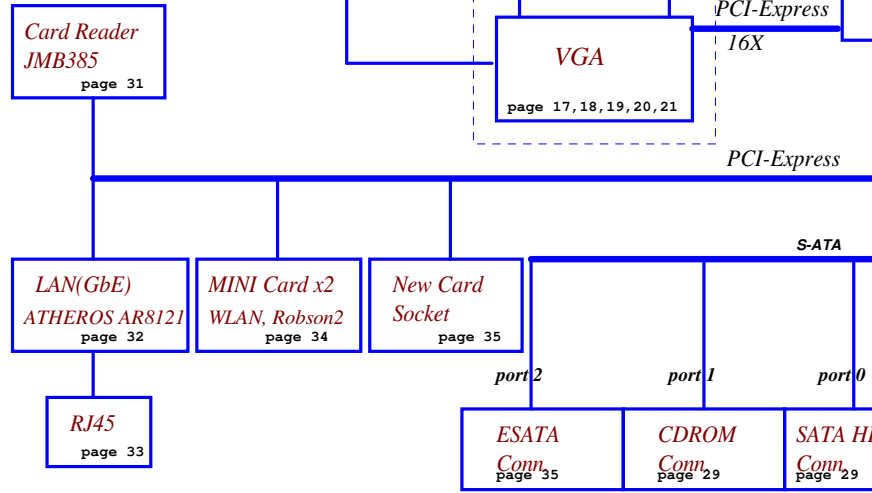
FSB
667/800/1066MHz
H_A#(3..35) H_D#(0..63)



Intel Cantiga
uFCBGA-1329
page 7, 8, 9, 10, 11, 12, 13

Memory BUS(DDRIII)
Dual Channel
1.5V DDRIII 800/1066
204pin DDRIII-SO-DIMM X2
BANK 0, 1, 2, 3 page 13, 14

Intel ICH9-M
BGA-676
page 25, 26, 27, 28



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Voltage Rails

Power Plane	Description	S1	S3	S5
VIN	Adapter power supply (19V)	N/A	N/A	N/A
B+	AC or battery power rail for power circuit.	N/A	N/A	N/A
+CPU_CORE	Core voltage for CPU	ON	OFF	OFF
+0.75VS	0.75V switched power rail for DDR terminator	ON	OFF	OFF
+1.05VS	1.05V switched power rail	ON	OFF	OFF
+1.25VS	1.25V switched power rail	ON	OFF	OFF
+1.5V	1.5V power rail for HDA/DDR3	ON	ON	OFF
+1.5VS	1.5V switched power rail	ON	OFF	OFF
+1.8V	1.8V GM LVDS MODULE	ON	ON	OFF
+1.8VS	1.8V switched power rail	ON	OFF	OFF
+1.1VS	1.1V switched power rail	ON	OFF	OFF
+3VALW	3.3V always on power rail	ON	ON	ON*
+3V	3.3V power rail for SB	ON	ON	X
+3V_LAN	3.3V power rail for LAN	ON	ON	X
+3VS	3.3V switched power rail	ON	OFF	OFF
+5VALW	5V always on power rail	ON	ON	ON*
+5VS	5V switched power rail	ON	OFF	OFF
+VSB	VSb always on power rail	ON	ON	ON*
+RTCVCC	RTC power	ON	ON	ON
+VGA_CORE	Core voltage for GPU	ON	OFF	OFF

Note : ON* means that this power plane is ON only with AC power available, otherwise it is OFF.

External PCI Devices

Device	IDSEL#	REQ#/GNT#	Interrupts
--------	--------	-----------	------------

EC SM Bus1 address

Device	Address	Device	Address
Smart Battery	0001 011X b	ADI ADT7421	1001 100X b
MEDIA CONSOLE	1010 000X b	NB9M THERMAL SENSOR	

EC SM Bus2 address

ICH9M SM Bus address

Device	Address
Clock Generator (ICS9LPRS387, SLG6SP556V)	1101 001Xb
DDR DIMM0	1001 000Xb
DDR DIMM2	1001 010Xb

BOM Configuration Table

Project	BOM Configuration
KAL90-UMA	XXXXXXXXXX: KAL90@/GM@/888VC@/8121@/GM45@
KAL90-Dis	XXXXXXXXXX: KAL90@/PM@/888VC@/8121@
KALH0-GM45	XXXXXXXXXX: KALH0@/GM@/888VC@/8121@/GM45@
KALH0-GL40	XXXXXXXXXX: KALH0@/GM@/888VC@/8121@/GL40@
KALH0-PM45	XXXXXXXXXX: KALH0@/PM@/888VC@/8121@
KAL90+ -UMA	GM@/888VC@/8121@/GM45@/KAL90+_G0@/KAL90_90+@/KAL90_G0_90+@/KAL90_H0_90+@/KAL90+_PCB@
KAL90+ -Dis	PM@/888VC@/8121@/KAL90+_G0@/KAL90_90+@/KAL90_G0_90+@/KAL90_H0_90+@/KAL90+_PCB@/PM45@
KALG0 -UMA (GL40)	KALG0@/GM@/888VC@/8121@/GL40@/KAL90+_G0@/KALH0_G0@/KAL90_G0_90+@/KALG0_DDR2 PCB RV0 @/KALG0+@
KALG0 -Dis	KALG0@/PM@/888VC@/8121@/PM45@/KAL90+_G0@/KALH0_G0@/KAL90_G0_90+@/KALG0_DDR2 PCB RV0 @/KALG0+@
KALG0 -UMA (GM45)	KALG0@/GM@/888VC@/8121@/GM45@/KAL90+_G0@/KALH0_G0@/KAL90_G0_90+@/KALG0_DDR2 PCB RV0 @/KALG0+@
KALG0 -DIS (GM45)	KALG0@/PM@/888VC@/8121@/GM45@/KAL90+_G0@/KALH0_G0@/KAL90_G0_90+@/KALG0_DDR2 PCB RV0 @/KALG0+@

KALG0 LAN to AR-8131----- 8121@ Change to 8131@

STATE	SIGNAL	SLP_S1#	SLP_S3#	SLP_S4#	SLP_S5#	+VALW	+V	+VS	Clock
Full ON		HIGH	HIGH	HIGH	HIGH	ON	ON	ON	ON
S1 (Power On Suspend)		LOW	HIGH	HIGH	HIGH	ON	ON	ON	LOW
S3 (Suspend to RAM)		LOW	LOW	HIGH	HIGH	ON	ON	OFF	OFF
S4 (Suspend to Disk)		LOW	LOW	LOW	HIGH	ON	OFF	OFF	OFF
S5 (Soft OFF)		LOW	LOW	LOW	LOW	ON	OFF	OFF	OFF

Board ID / SKU ID Table for AD channel

Vcc	3.3V +/- 5%			
Ra/Rc/Re	100K / Rd / - 5%			
Board ID	Rb / Rd / Rf	VAD_BID min	VAD_BID typ	VAD_BID max
0	0	0 V	0 V	0 V
1	8.2K +/- 5%	0.216 V	0.250 V	0.289 V
2	18K +/- 5%	0.436 V	0.503 V	0.538 V
3	33K +/- 5%	0.712 V	0.819 V	0.875 V
4	56K +/- 5%	1.036 V	1.185 V	1.264 V
5	100K +/- 5%	1.453 V	1.650 V	1.759 V
6	200K +/- 5%	1.935 V	2.200 V	2.341 V
7	NC	2.500 V	3.300 V	3.300 V

BOARD ID Table

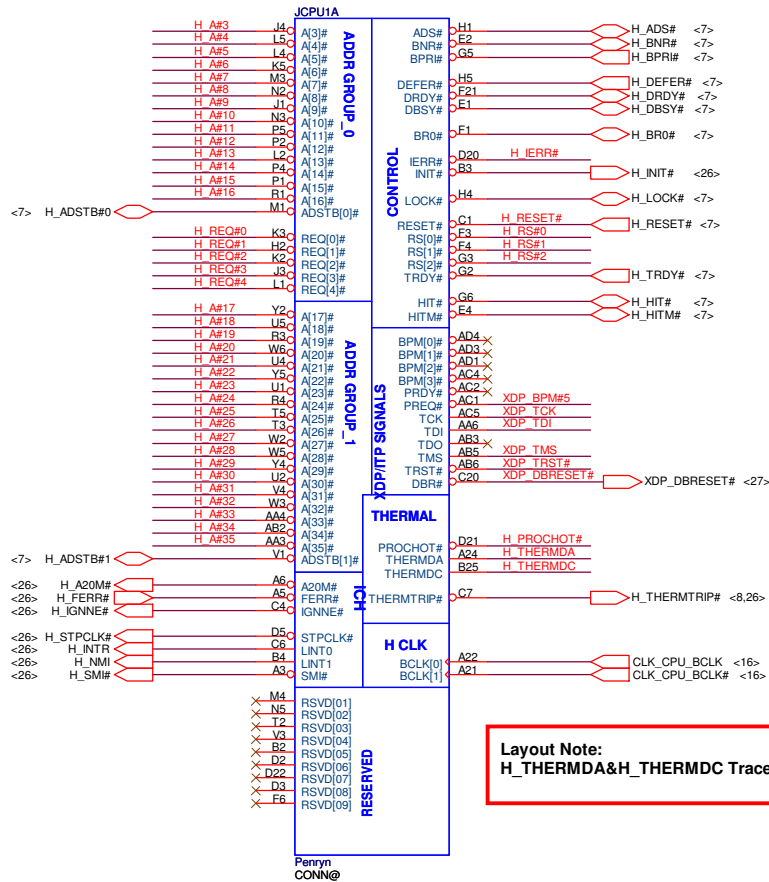
Board ID	PCB Revision
0	0.1
1	0.2
2	0.3
3	1.0
4	1A
5	
6	
7	

BTO Option Table

BTO Item	BOM Structure
KAL90	KAL90@
UMA	GM@
PM@	PM@
ALC888VC	888VC@
ALC888VB	888VB@
AR8121	8121@
AR8112	8112@
ALC268	268@
GL40	GL40@
GM45	GM45@
KAL90-G0	KAL90_G0@
KAL90-H0	KAL90_H0@
KALG0	KALG0@
KALH0	KALH0@
ALC268	268@
	KAL90_90+@
	KAL90_H0_G0@
	KAL90+_G0
	KALH0_G0
	KAL90_G0_90+@
	KAL90_H0_90+@
	KAL90+_PCB@
	KALG0_PCB@

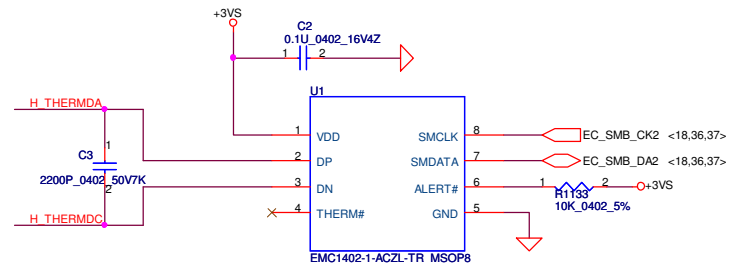
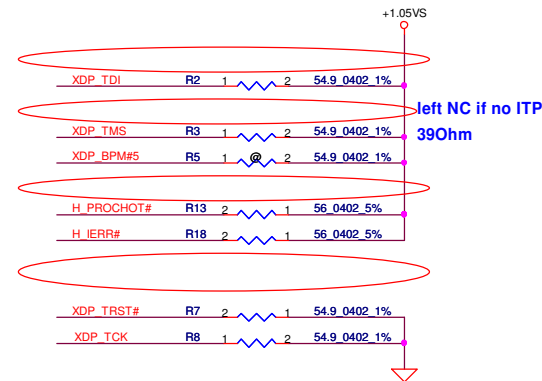
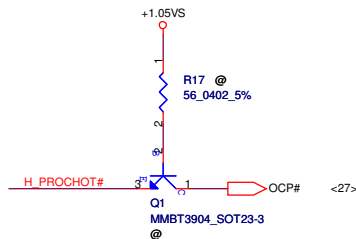
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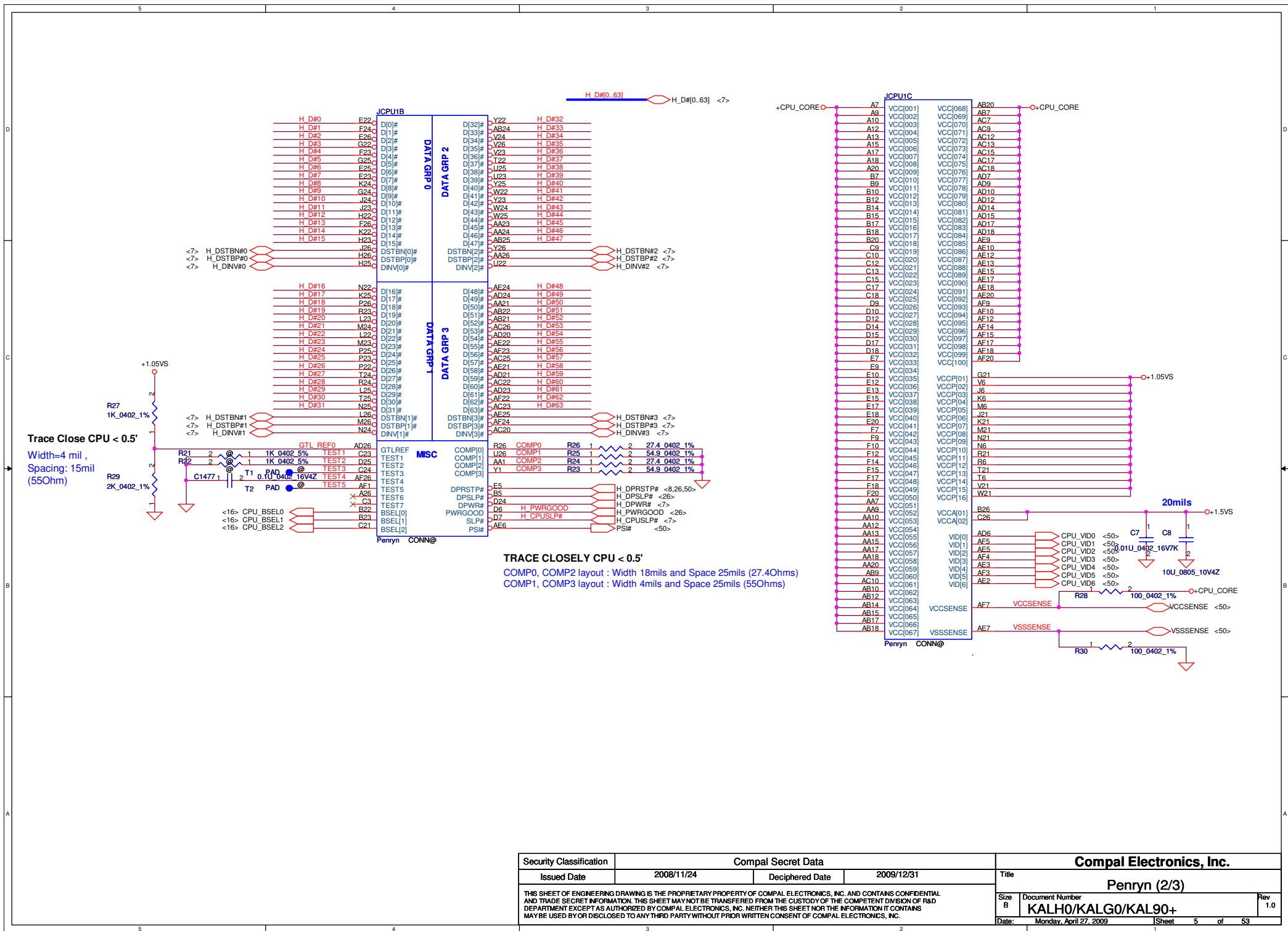
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 <7> H_RS# [0..2]  H_RS# [0..2]

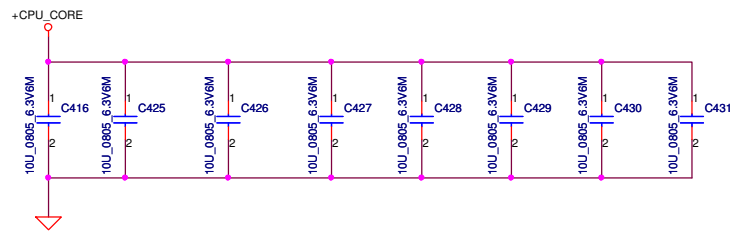
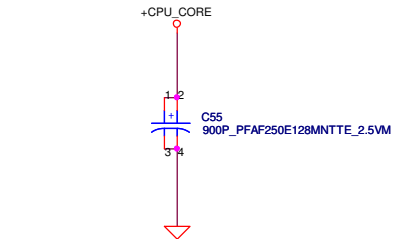
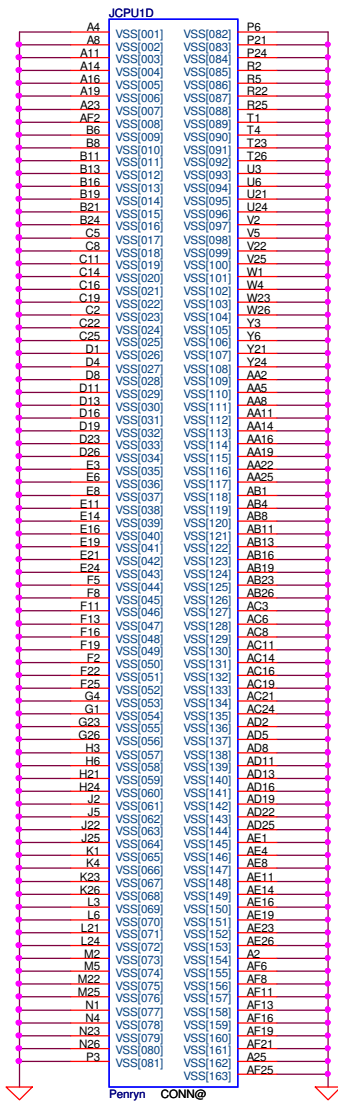


Layout Note:
 H_THERMDA & H_THERMDC Trace / Space = 10 / 10 mil

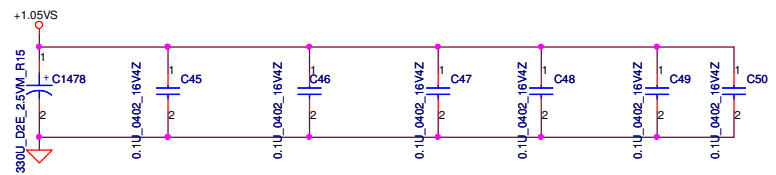
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0	0	0	266
0	1	0	200
0	1	1	166

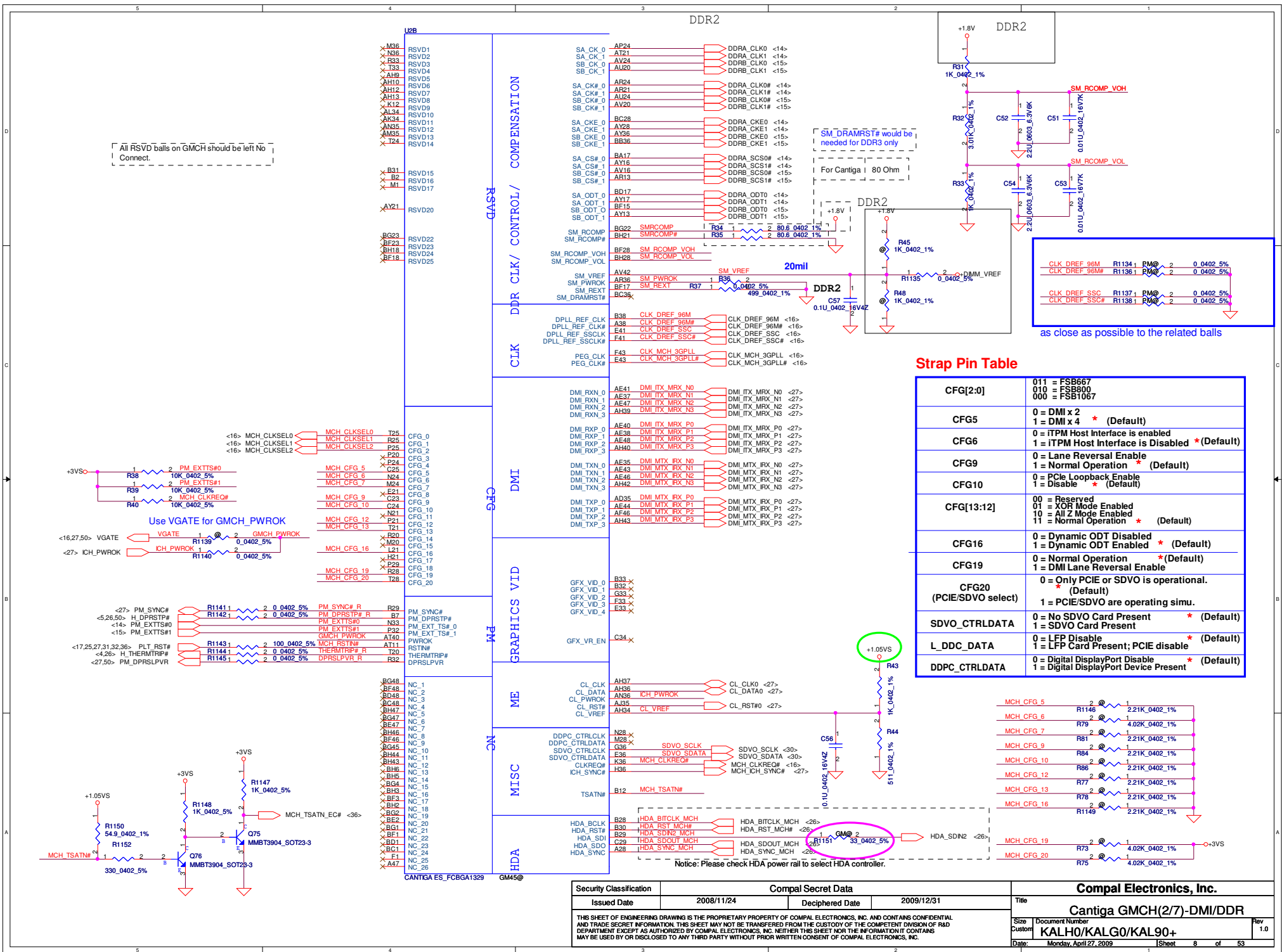






+CPU-CORE Decoupling	C,uF	ESR, mohm	ESL,nH
SPCAP,Polymer	4X330uF	6m ohm/4	1.8nH/6
MLCC 0805 X5R	32X22uF	3m ohm/32	0.6nH/32
	32X10uF	3m ohm/32	0.6nH/32





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DDRA_SDQ1 AJ41 SA_DQ_1
DDRA_SDQ2 AN38 SA_DQ_2
DDRA_SDQ3 AM38 SA_DQ_3
DDRA_SDQ4 AJ36 SA_DQ_4
DDRA_SDQ5 AJ40 SA_DQ_5
DDRA_SDQ6 AM44 SA_DQ_6
DDRA_SDQ7 AM42 SA_DQ_7
DDRA_SDQ8 AN43 SA_DQ_8
DDRA_SDQ9 AN44 SA_DQ_9
DDRA_SDQ10 AU40 SA_DQ_10
DDRA_SDQ11 AT38 SA_DQ_11
DDRA_SDQ12 AN41 SA_DQ_12
DDRA_SDQ13 AN39 SA_DQ_13
DDRA_SDQ14 AU44 SA_DQ_14
DDRA_SDQ15 AU42 SA_DQ_15
DDRA_SDQ16 AY39 SA_DQ_16
DDRA_SDQ17 AY44 SA_DQ_17
DDRA_SDQ18 BA40 SA_DQ_18
DDRA_SDQ19 BD43 SA_DQ_19
DDRA_SDQ20 AV41 SA_DQ_20
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DDRA_SDQ22 BB41 SA_DQ_22
DDRA_SDQ23 BC40 SA_DQ_23
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DDRA_SDQ26 AY37 SA_DQ_26
DDRA_SDQ27 AT36 SA_DQ_27
DDRA_SDQ28 AY38 SA_DQ_28
DDRA_SDQ29 BB38 SA_DQ_29
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DDRA_SDQ50 AT9 SA_DQ_50
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DDR SYSTEM MEMORY A

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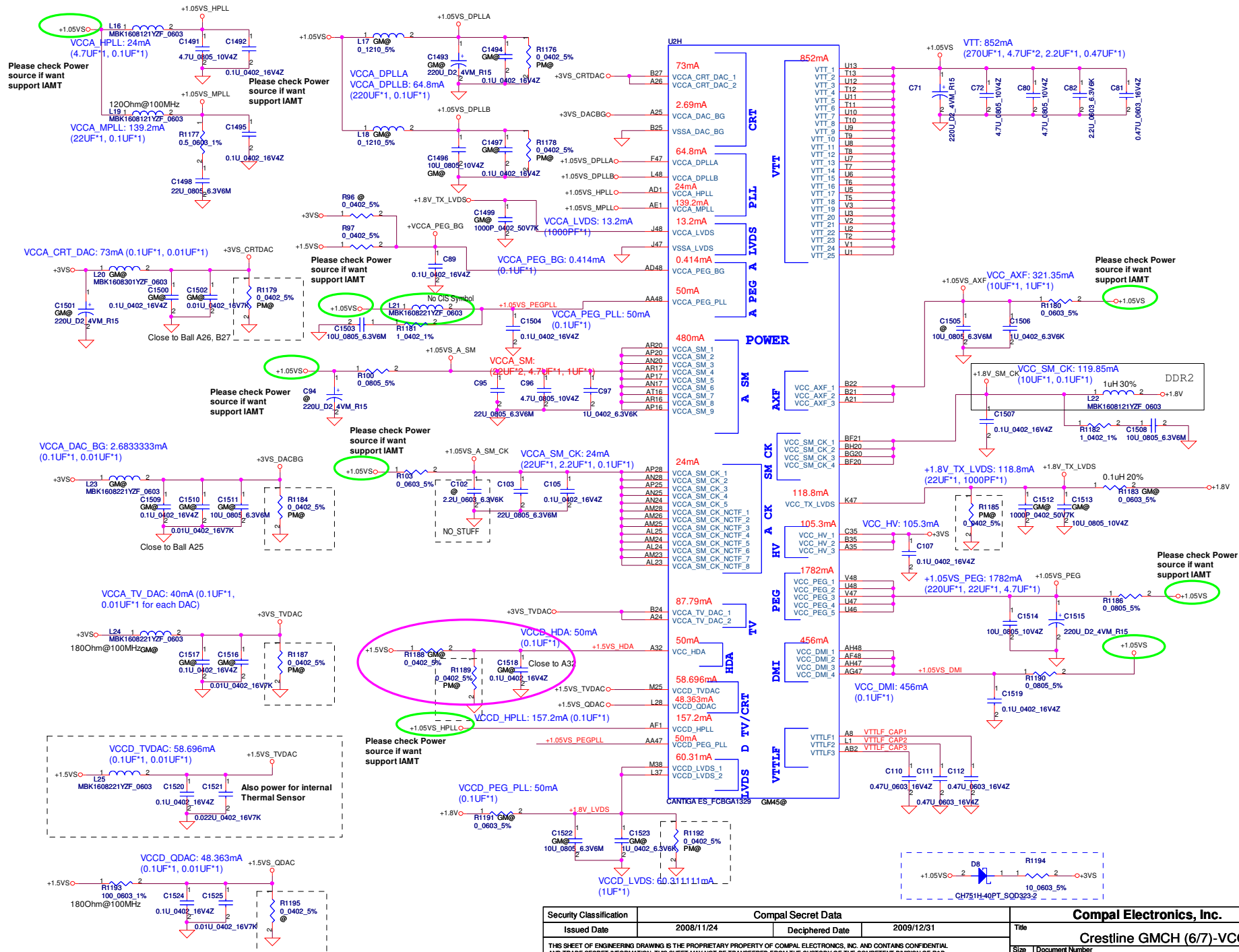
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BD24 DDRA_SMA6
BG27 DDRA_SMA7
BF25 DDRA_SMA8
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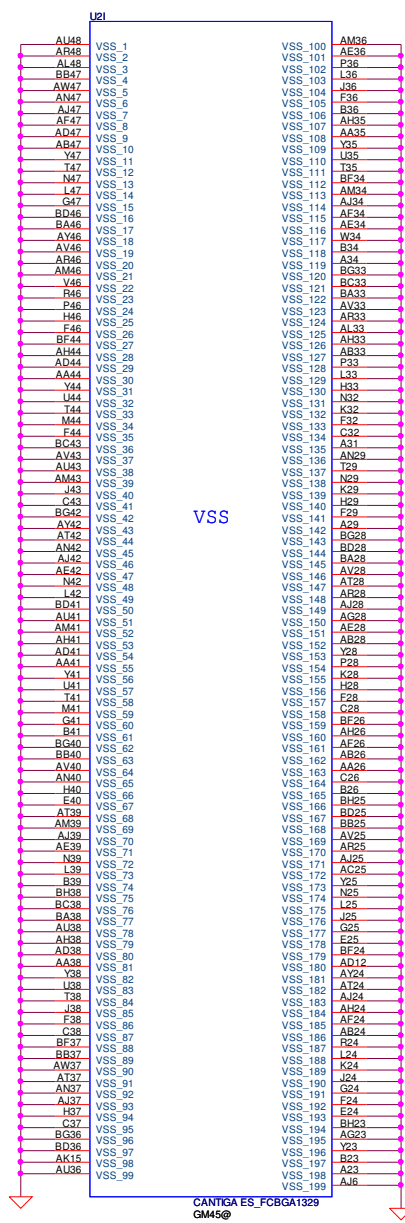
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DDRB_SDQ45 BF5 SB_DQ_45
DDRB_SDQ46 BA1 SB_DQ_46
DDRB_SDQ47 BD3 SB_DQ_47
DDRB_SDQ48 AV2 SB_DQ_48
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DDR SYSTEM MEMORY B

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GM45@

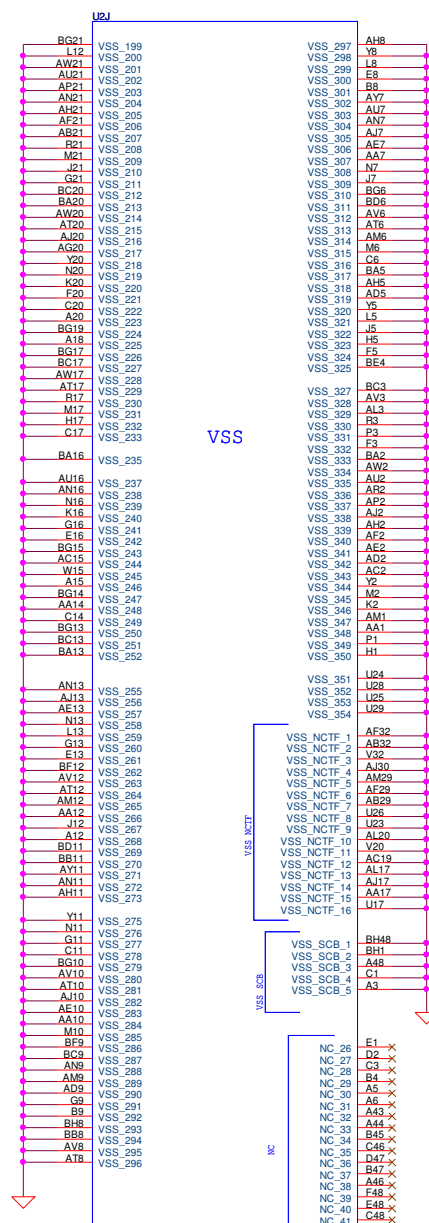
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BF35 DDRB_SDM3
BG11 DDRB_SDM4
BA3 DDRB_SDM5
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AV17 DDRB_SMA0
BA25 DDRB_SMA1
BC25 DDRB_SMA2
AU25 DDRB_SMA3
AV25 DDRB_SMA4
BB28 DDRB_SMA5
AU28 DDRB_SMA6
AT28 DDRB_SMA7
AT33 DDRB_SMA8
BD33 DDRB_SMA9
BB16 DDRB_SMA10
AW33 DDRB_SMA11
AY33 DDRB_SMA12
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AU33 DDRB_SMA14





VSS

CANTIGA ES_FCBGA1329
GM45@



VSS

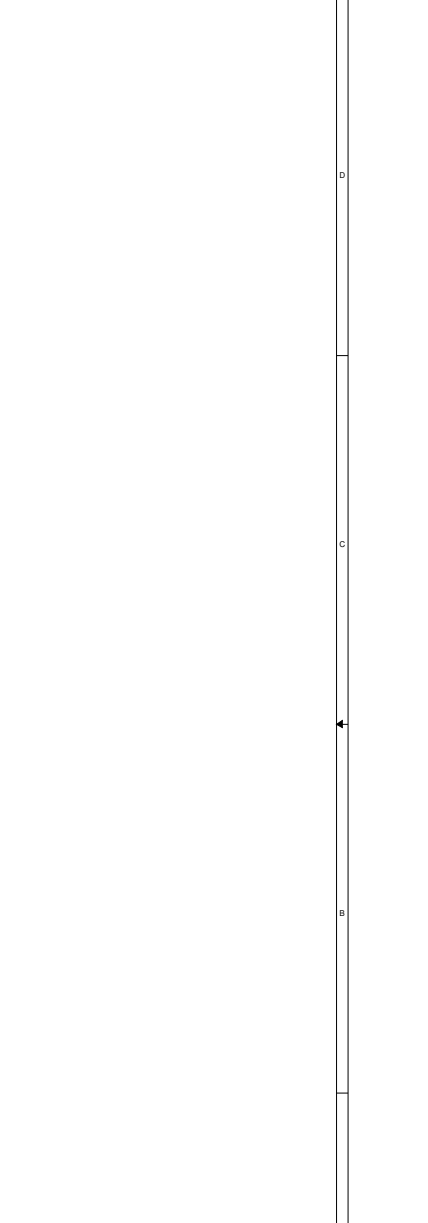
CANTIGA ES_FCBGA1329
GM45@



VSS NCTF

VSS SCB

NC



Compal Secret Data

2008/11/24

Deciphered Date

2009/12/31

Title

Cantiga GMCH(1/7)-GTL

Size Document Number

Customer KALHO/KALGO/KAL90+

Date Monday, April 27, 2009

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Security Classification
Issued Date
2008/11/24
Deciphered Date
2009/12/31
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Compal Electronics, Inc.
Cantiga GMCH(1/7)-GTL
Size Document Number
Customer KALHO/KALGO/KAL90+
Date Monday, April 27, 2009
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DIMM2 REV H:10.1mm (BOT)

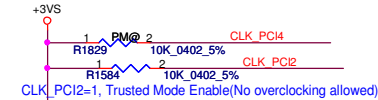
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Size B	Document Number	Rev		Date	
	KAL90	0.1		Monday, April 27, 2009	
Sheet		15		of 53	

FSLC CLKSEL2	FSLB CLKSEL1	FSLA CLKSEL0	CPU MHz	SRC MHz	PCI MHz
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0	1	0	200	100	33.3
0	1	1	166	100	33.3

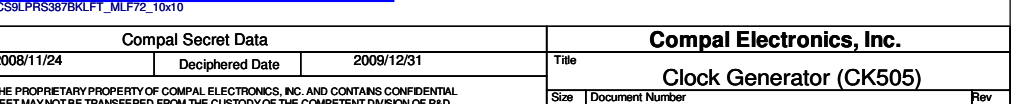
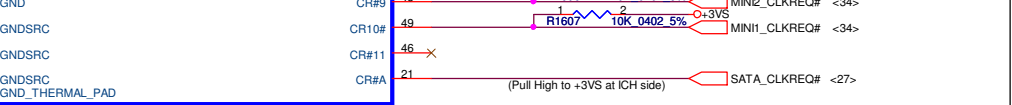
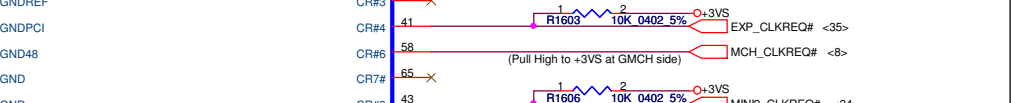
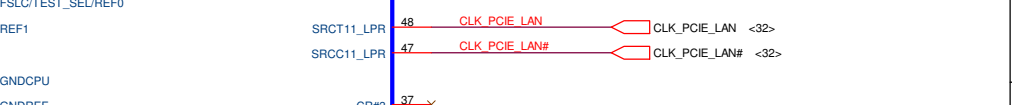
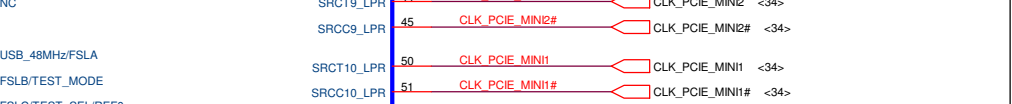
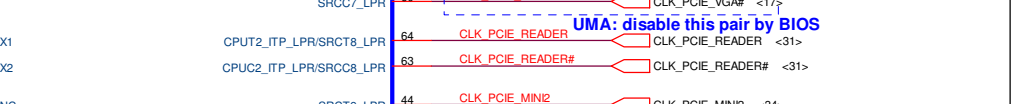
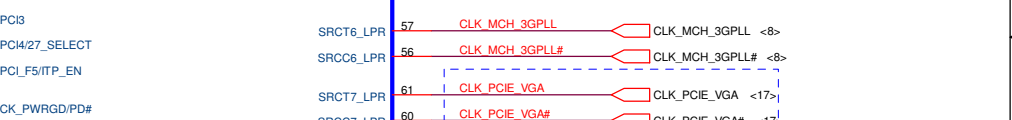
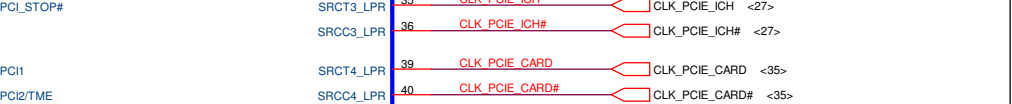
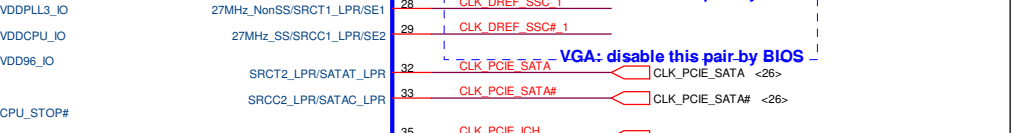
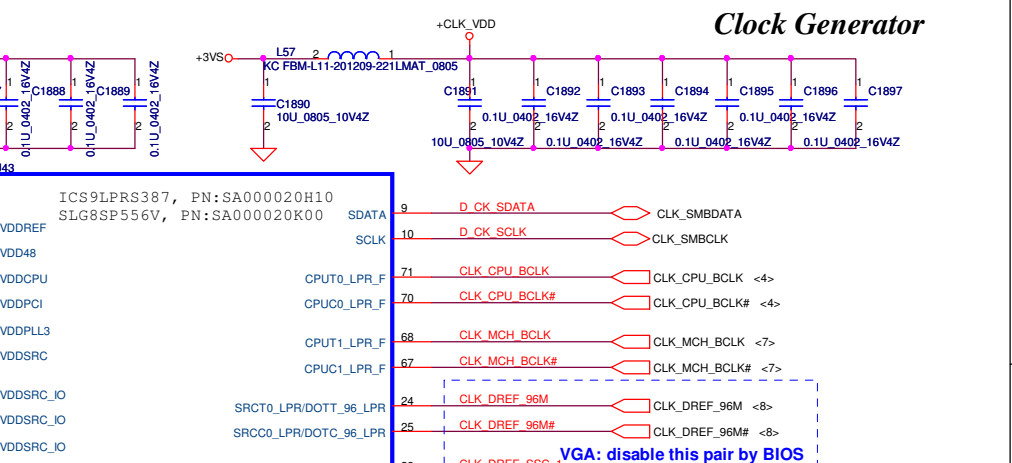
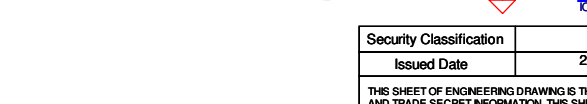
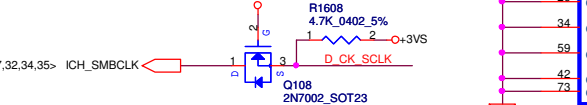
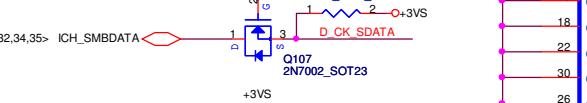
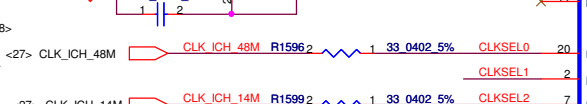
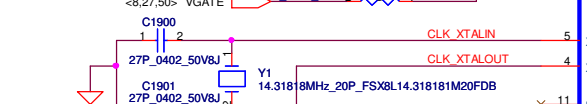
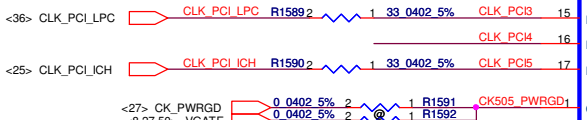
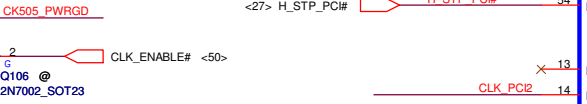
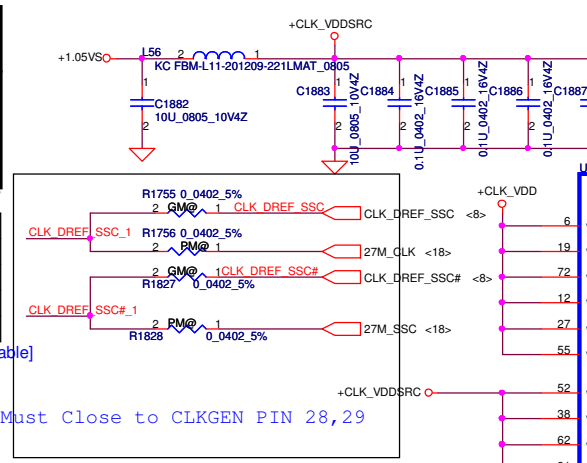
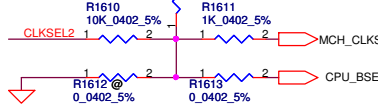
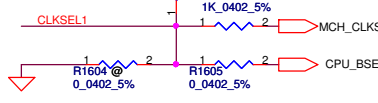
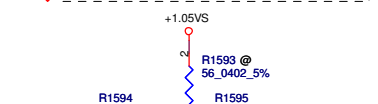
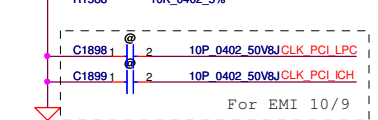
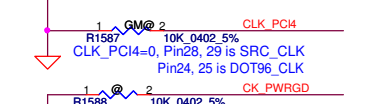
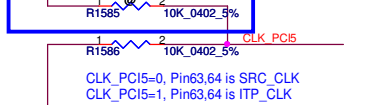
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CLK_REQ#	Control	Free-Run
CR#_10(WLAN)	PCIEX10	PCIEX0
CR#_6(MCH)	PCIEX6	PCIEX1
CR#_4(NEW CARD)	PCIEX4	
CR#_9(MINI CARDII)	PCIEX9	

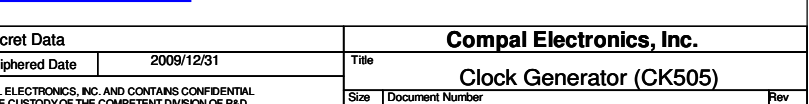
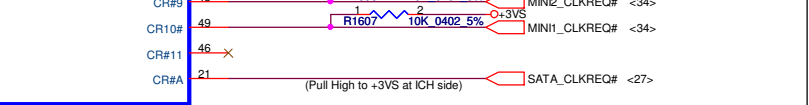
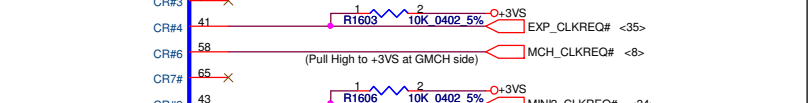
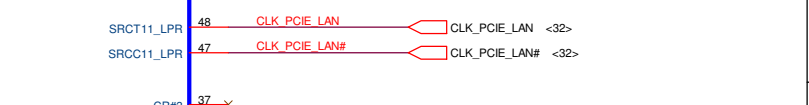
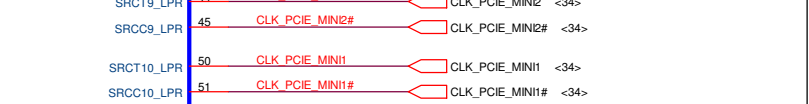
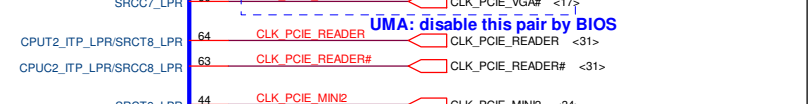
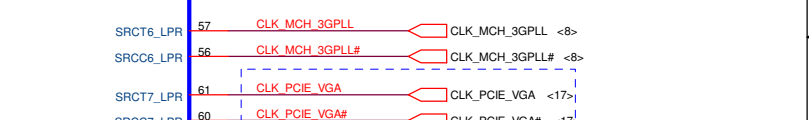
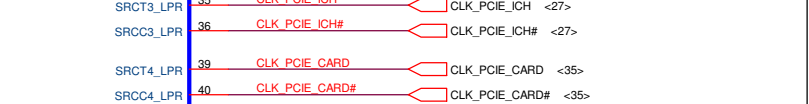
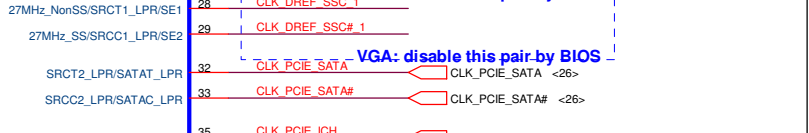
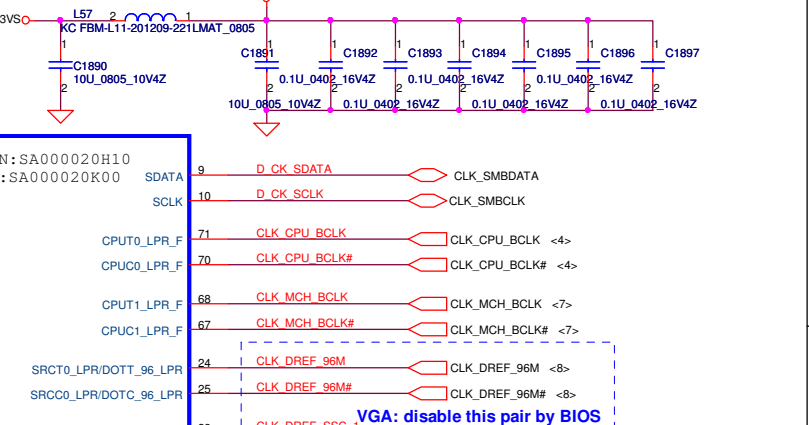
SRC7(VGA_CLK): Discrete VGA[Enable] UMA[Disable]



mount to Enable ITP_CLK



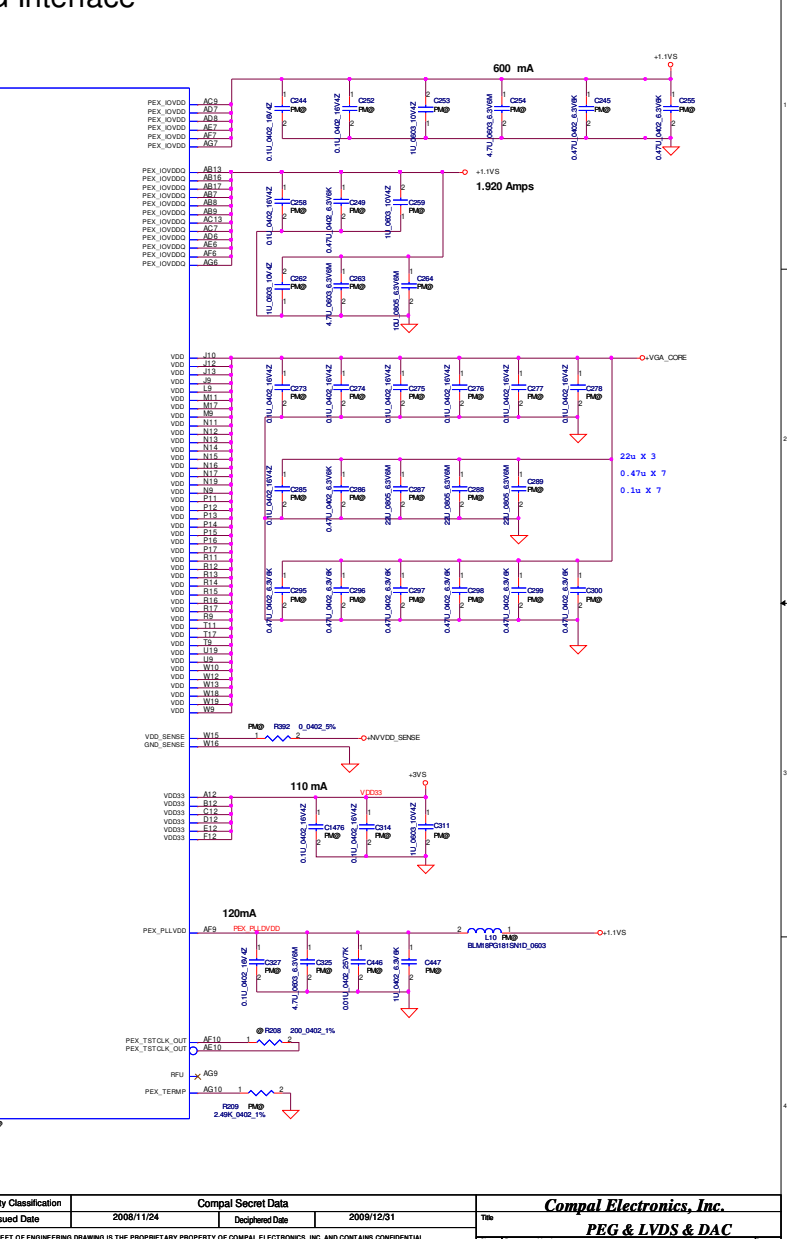
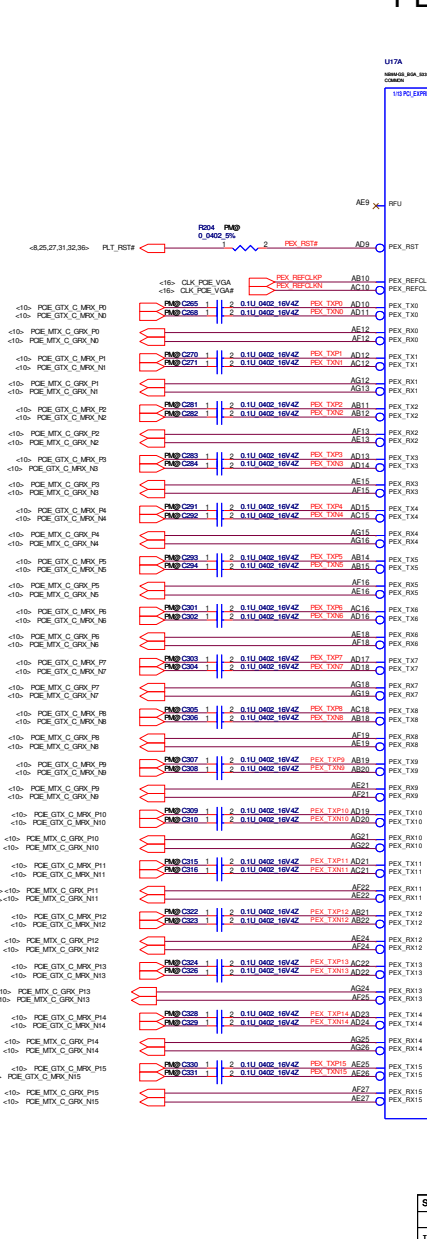
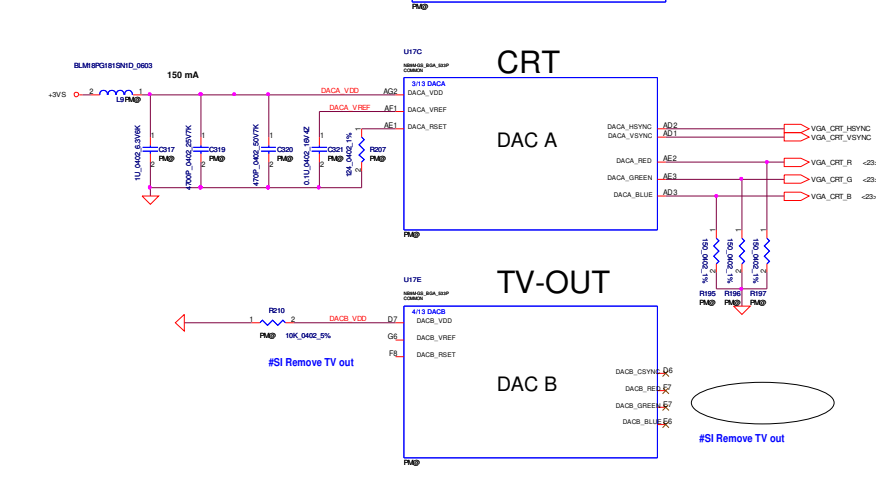
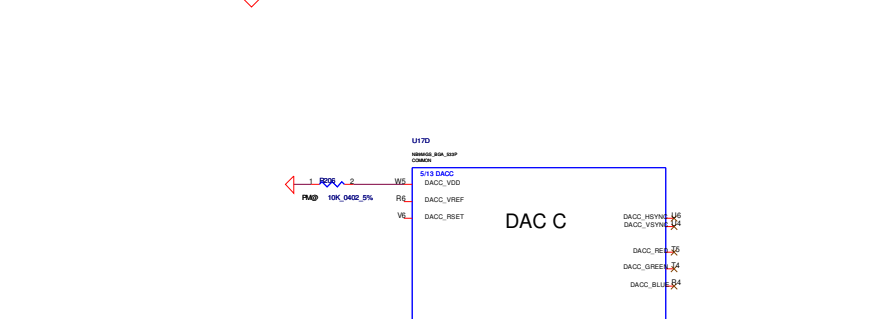
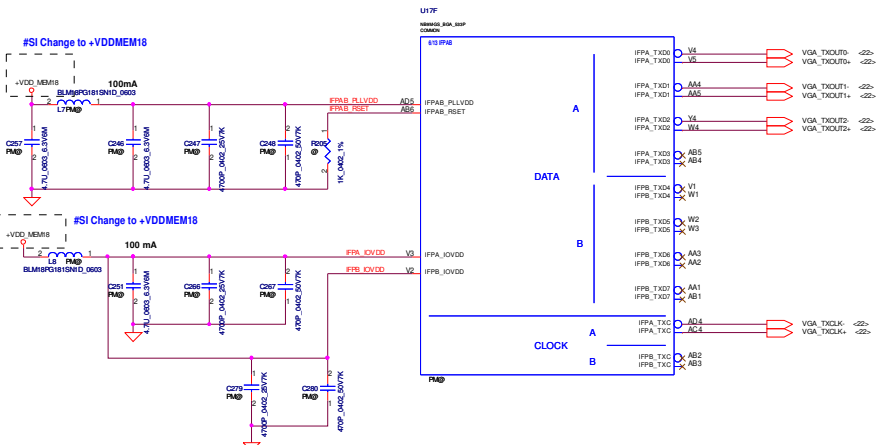
Clock Generator



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Issued Date	2008/11/24		Deciphered Date	2009/12/31		Title		
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				Size	Document Number		Rev	
				Custom	KALH0/KALG0/KAL90+		1.0	
Date:		Monday, April 27, 2009		Sheet		16 of 53		

LVDS & DAC Interface

PEG Interface



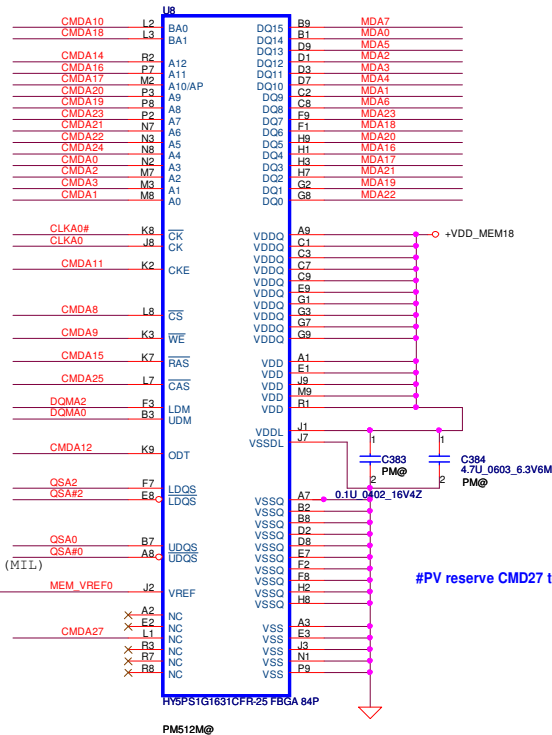
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VRAM DDR2 chips (256MB & 512MB)

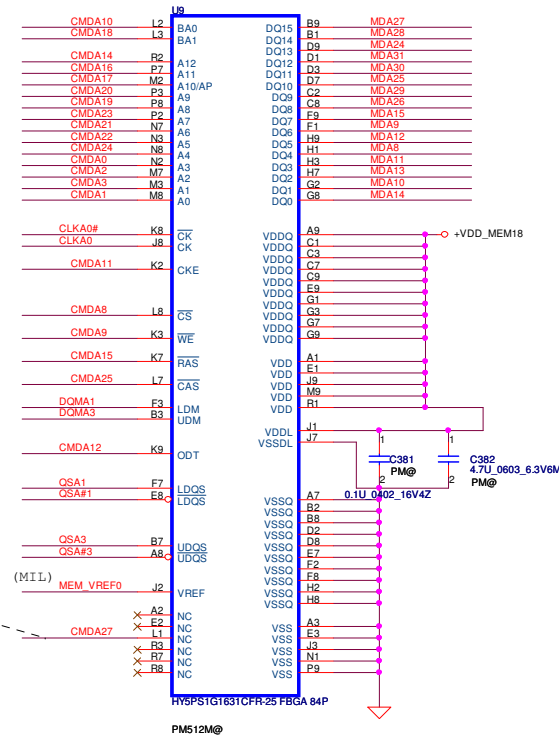
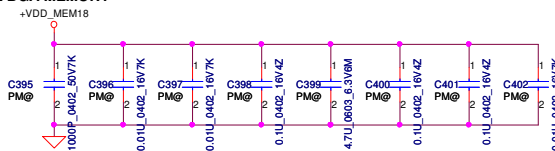
32Mx16 DDR2 400MHz *4==>256MB

64Mx16 DDR2 400MHz*4==>512MB

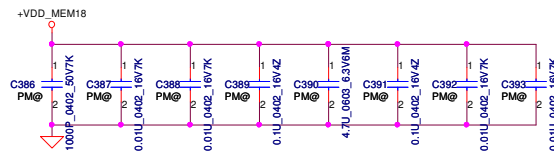
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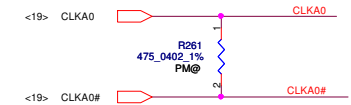
DDR2 BGA MEMORY



DDR BGA MEMORY



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CMD1	A0	A0
CMD2	A2	
CMD3	A1	A1
CMD4		A3
CMD5		A4
CMD6		A5
CMD7		
CMD8	CS#	CS#
CMD9	WE#	WE#
CMD10	BA0	BA0
CMD11	CKE	CKE
CMD12	ODT	ODT
CMD13		
CMD14	A12	A12
CMD15	RAS#	RAS#
CMD16	A11	A11
CMD17	A10	A10
CMD18	BA1	BA1
CMD19	A8	A8
CMD20	A9	A9
CMD21	A6	A6
CMD22	A5	
CMD23	A7	A7
CMD24	A4	
CMD25	CAS#	CAS#
CMD26	A13	A13
CMD27	BA2	BA2
CMD28		
CMD29		
CMD30		

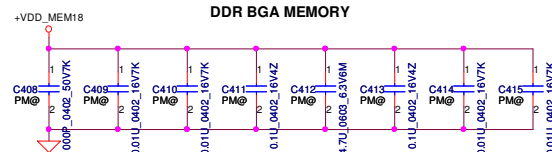
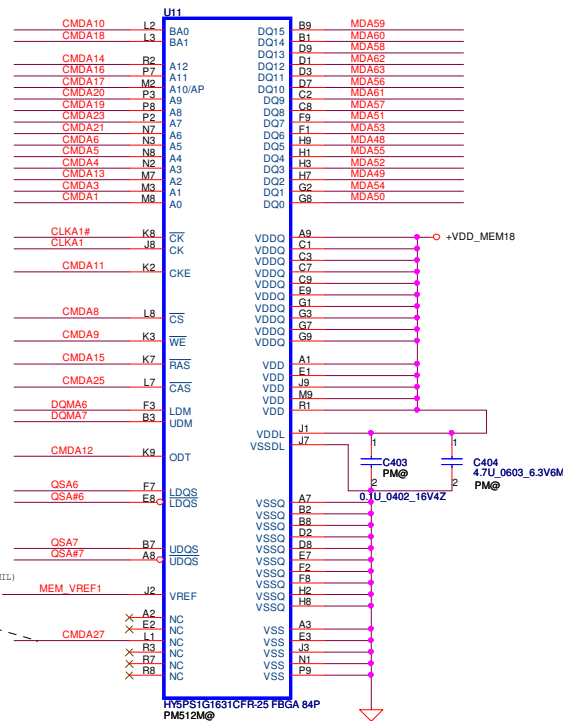
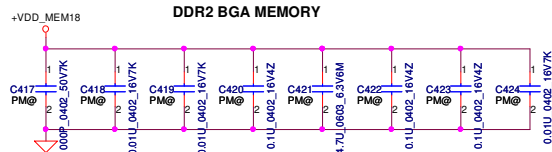
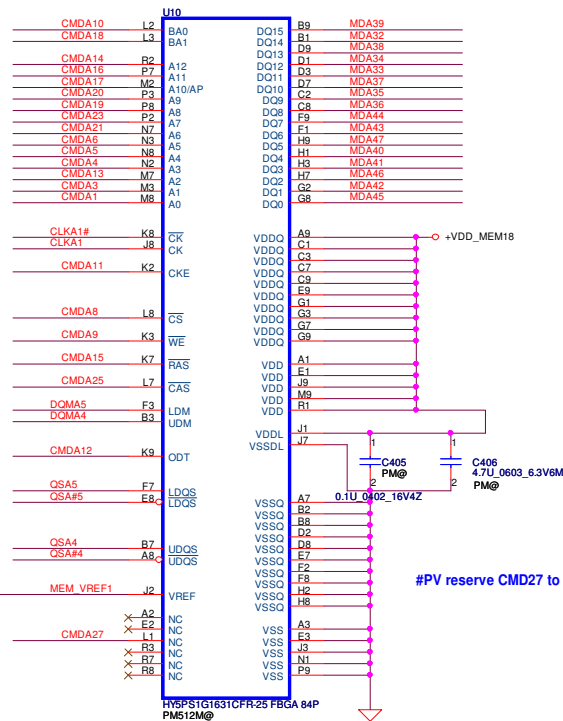
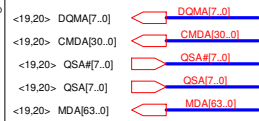


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Issued Date	2008/11/24	Deciphered Date	2009/12/31	VRAM 1	
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					KAL90
				Date:	Monday, April 27, 2009
				Sheet	7 of 16

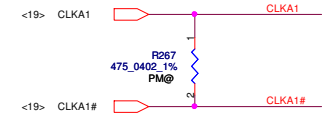
VRAM DDR2 chips (256MB & 512MB)

32Mx16 DDR2 400MHz *4==>256MB

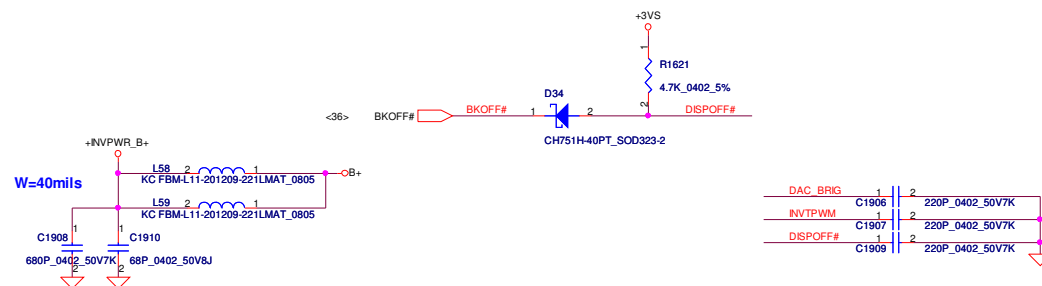
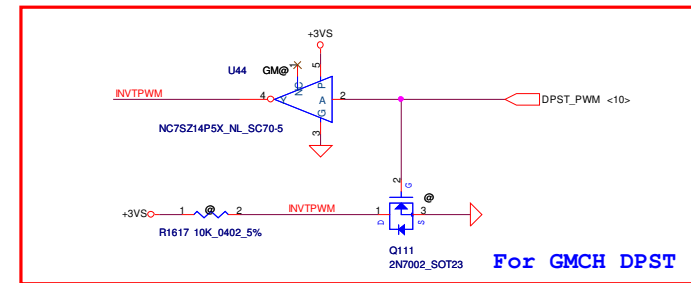
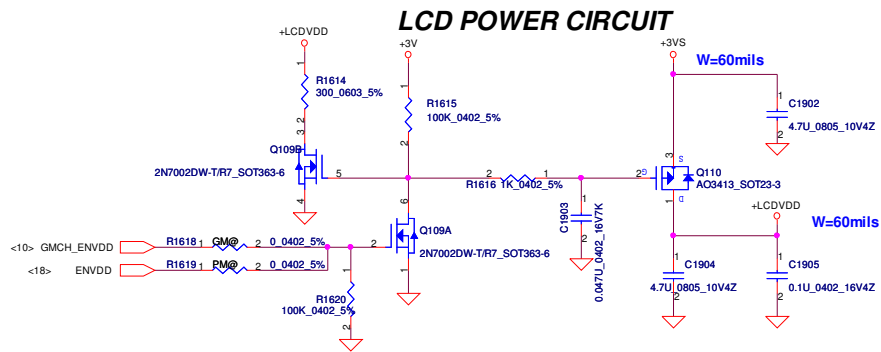
64Mx16 DDR2 400MHz*4==>512MB



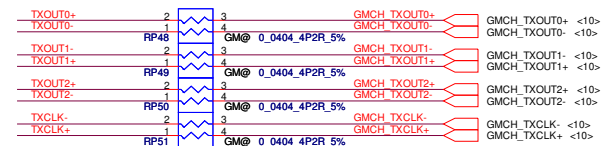
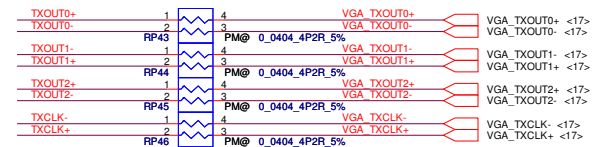
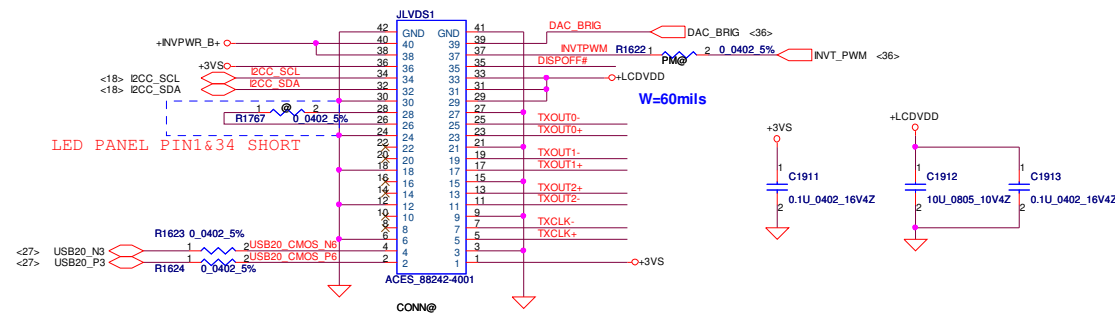
DATA Bus		
Address	0..31	32..63
CMD0	A3	
CMD1	A0	A0
CMD2	A2	
CMD3	A1	A1
CMD4		A3
CMD5		A4
CMD6		A5
CMD7		
CMD8	CS#	CS#
CMD9	WE#	WE#
CMD10	BA0	BA0
CMD11	CKE	CKE
CMD12	ODT	ODT
CMD13		
CMD14	A12	A12
CMD15	RAS#	RAS#
CMD16	A11	A11
CMD17	A10	A10
CMD18	BA1	BA1
CMD19	A8	A8
CMD20	A9	A9
CMD21	A6	A6
CMD22	A5	
CMD23	A7	A7
CMD24	A4	
CMD25	CAS#	CAS#
CMD26	A13	A13
CMD27	BA2	BA2
CMD28		
CMD29		
CMD30		



Security Classification		Compal Secret Data		Title	
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				KAL90	0.1
				Date: Monday, April 27, 2009	Sheet 8 of 16

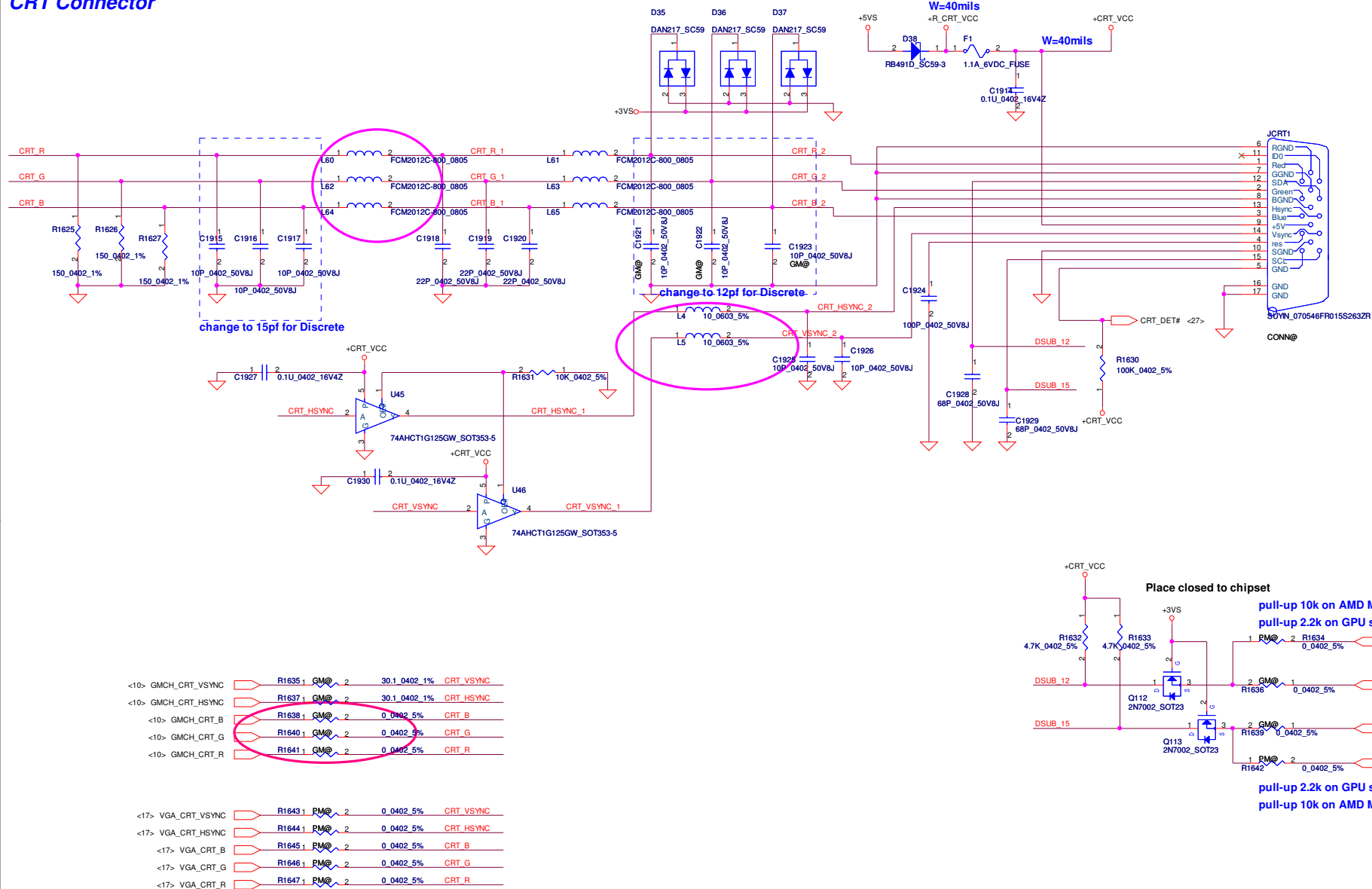


LCD/PANEL BD. Conn.

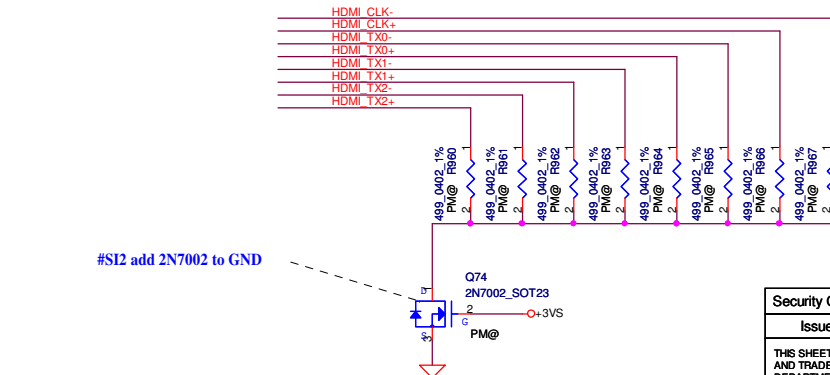
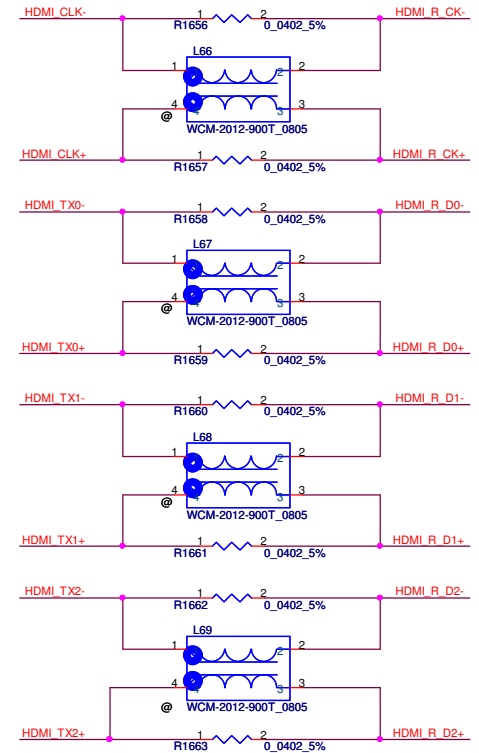
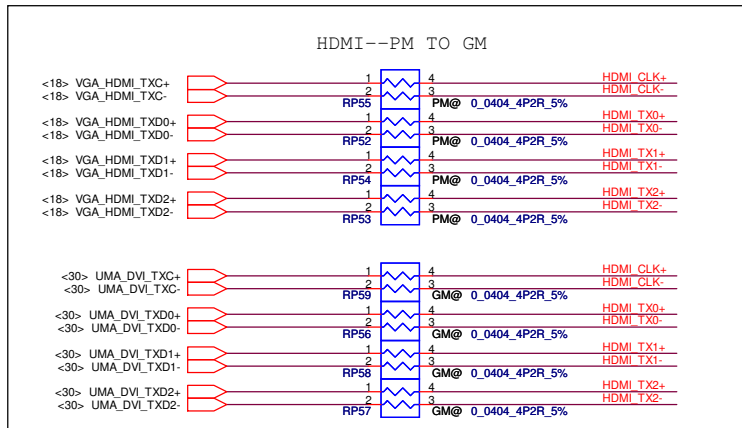
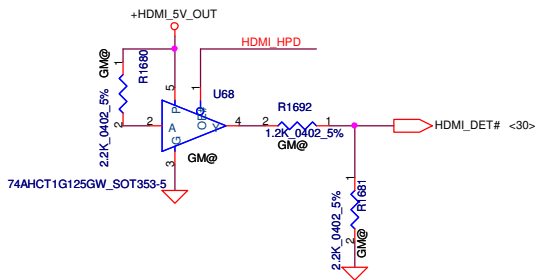
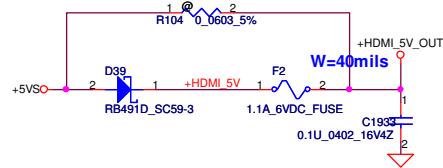
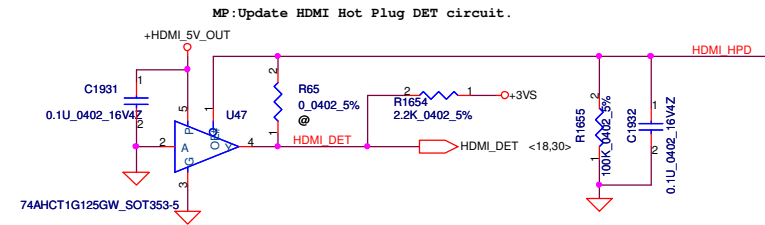
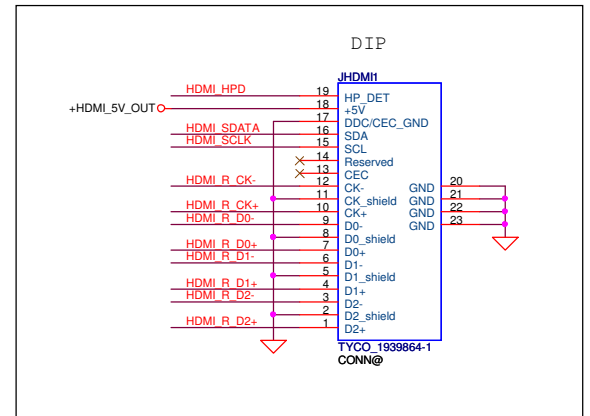
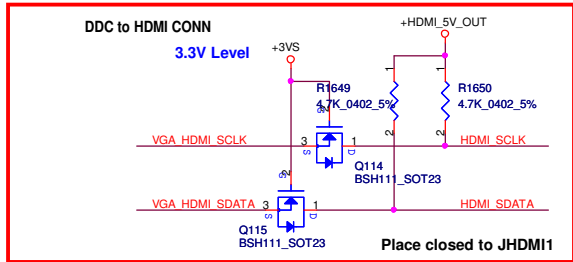
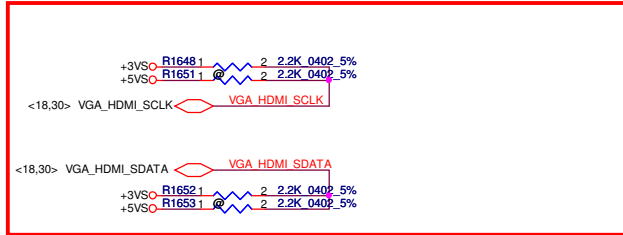


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Size	Document Number	Customer	Date	Monday, April 27, 2009	Sheet 22 of 53
KALHO/KALGO/KAL90+				Rev 1.0	

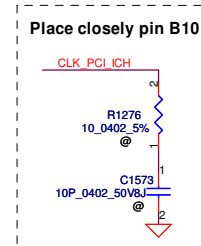
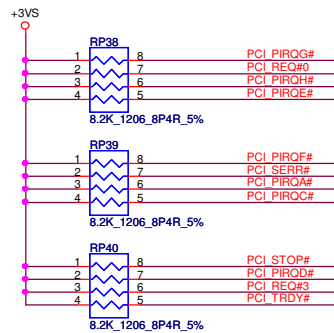
CRT Connector



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Size	B	Document Number	KALHO/KALGO/KAL90+		Rev
Date	Monday, April 27, 2009	Sheet	23	of	53



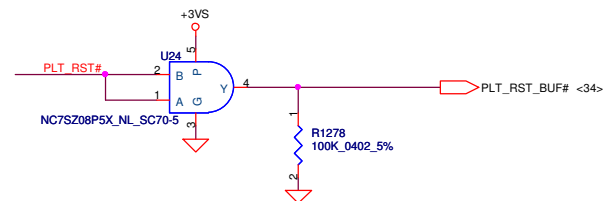
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Issued Date	2008/11/24	Deciphered Date	2009/12/31	Title	
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Size		Document Number		Rev	
Custom		KALH0/KALG0/KAL90+		1.0	
Date		Monday, April 27, 2009		Sheet 24 of 53	



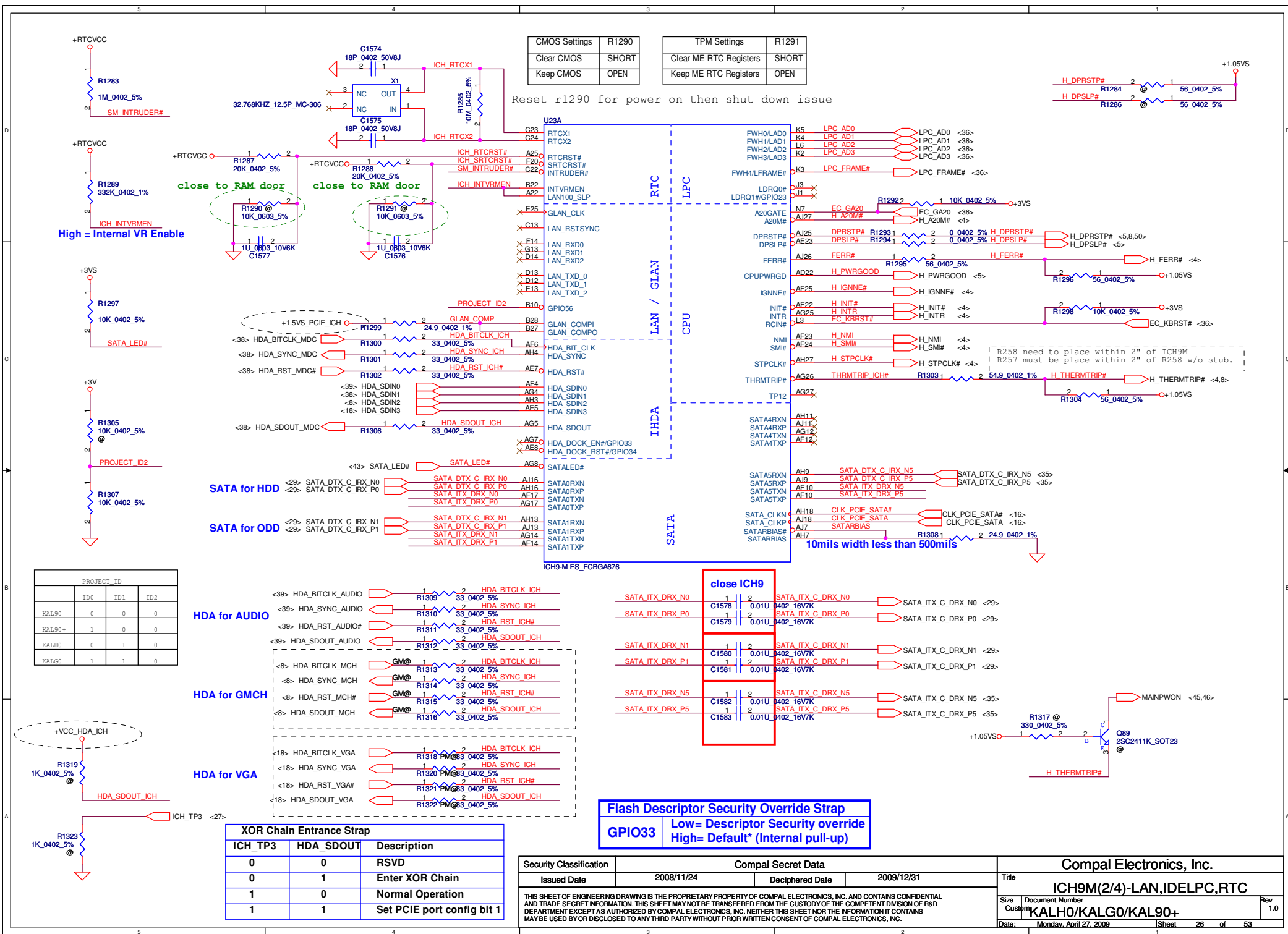
R1277 1  2 1K 0402 5% PCI GNT#3

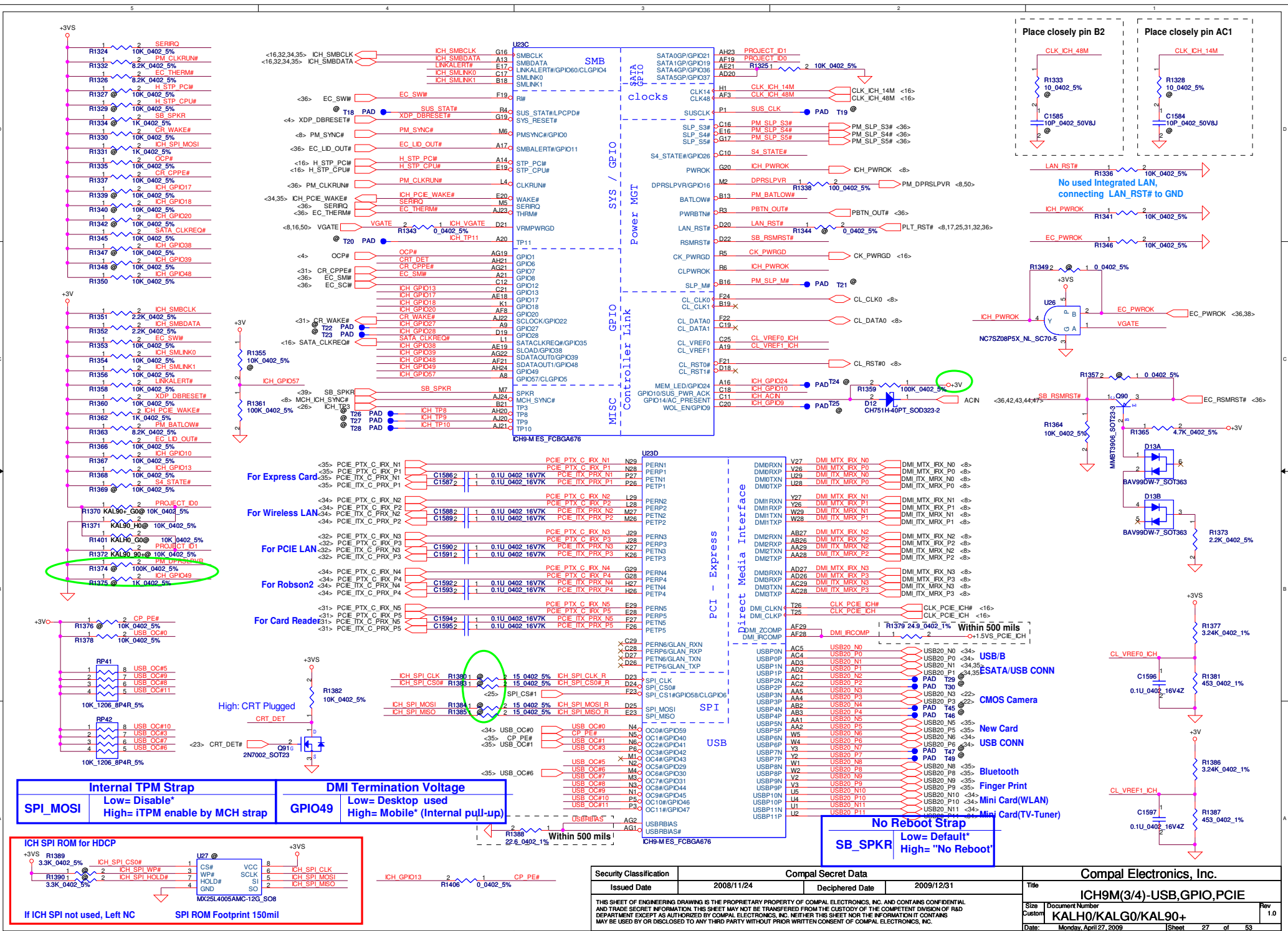
R1280 1 @ 2 1K 0402 5% PCI GNT#0

R1281 1 @ 2 1K 0402 5% SPL_CS#1 <27>

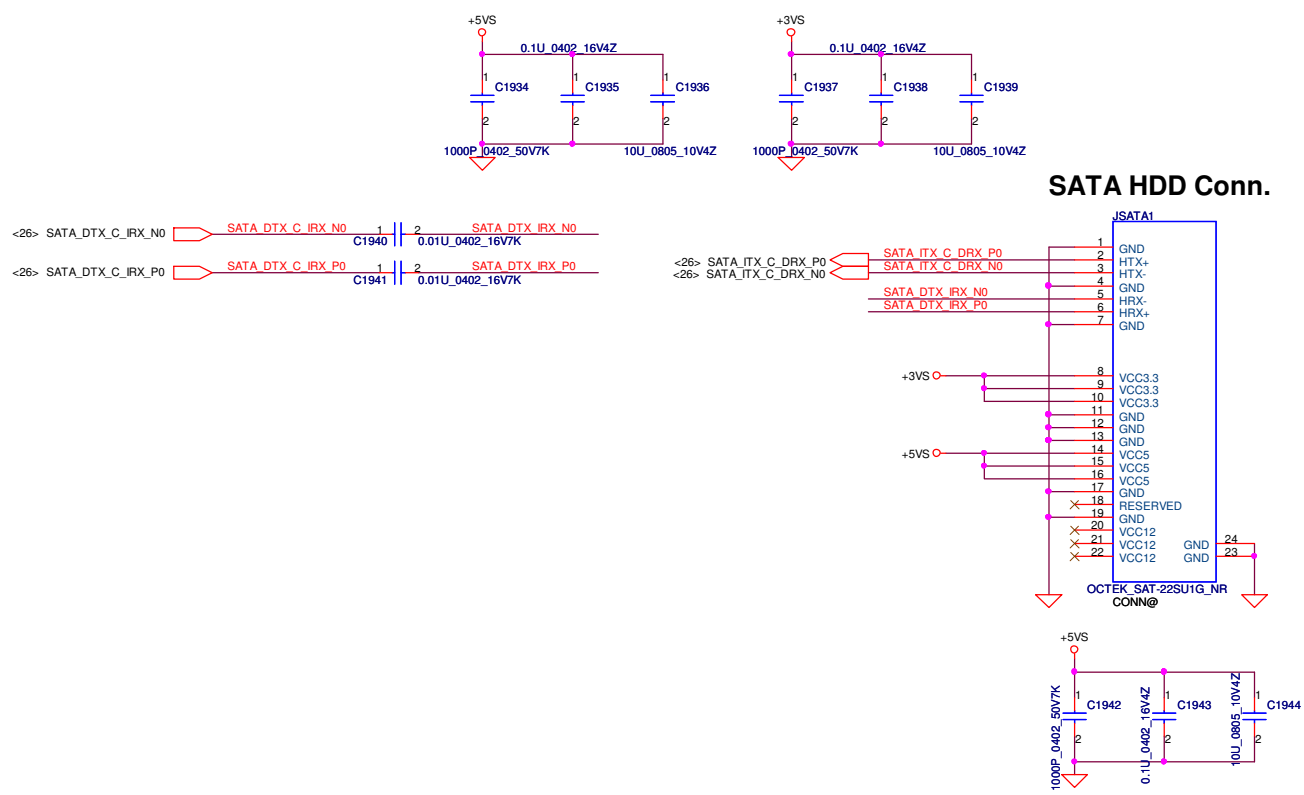


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				KALHO/KALGO/KAL90+		
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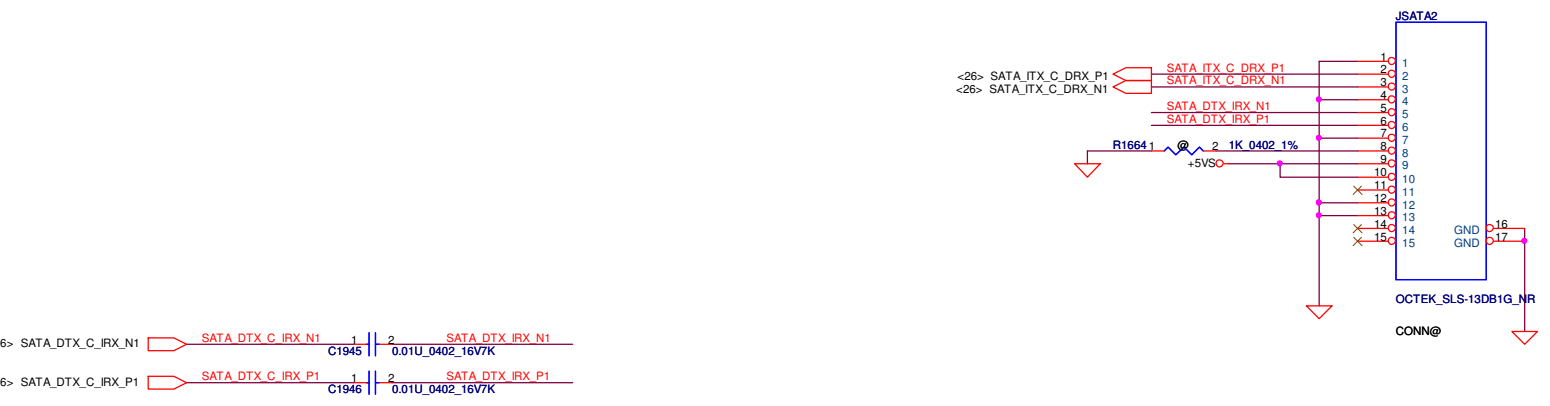


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SATA HDD Conn.

SATA ODD Conn.



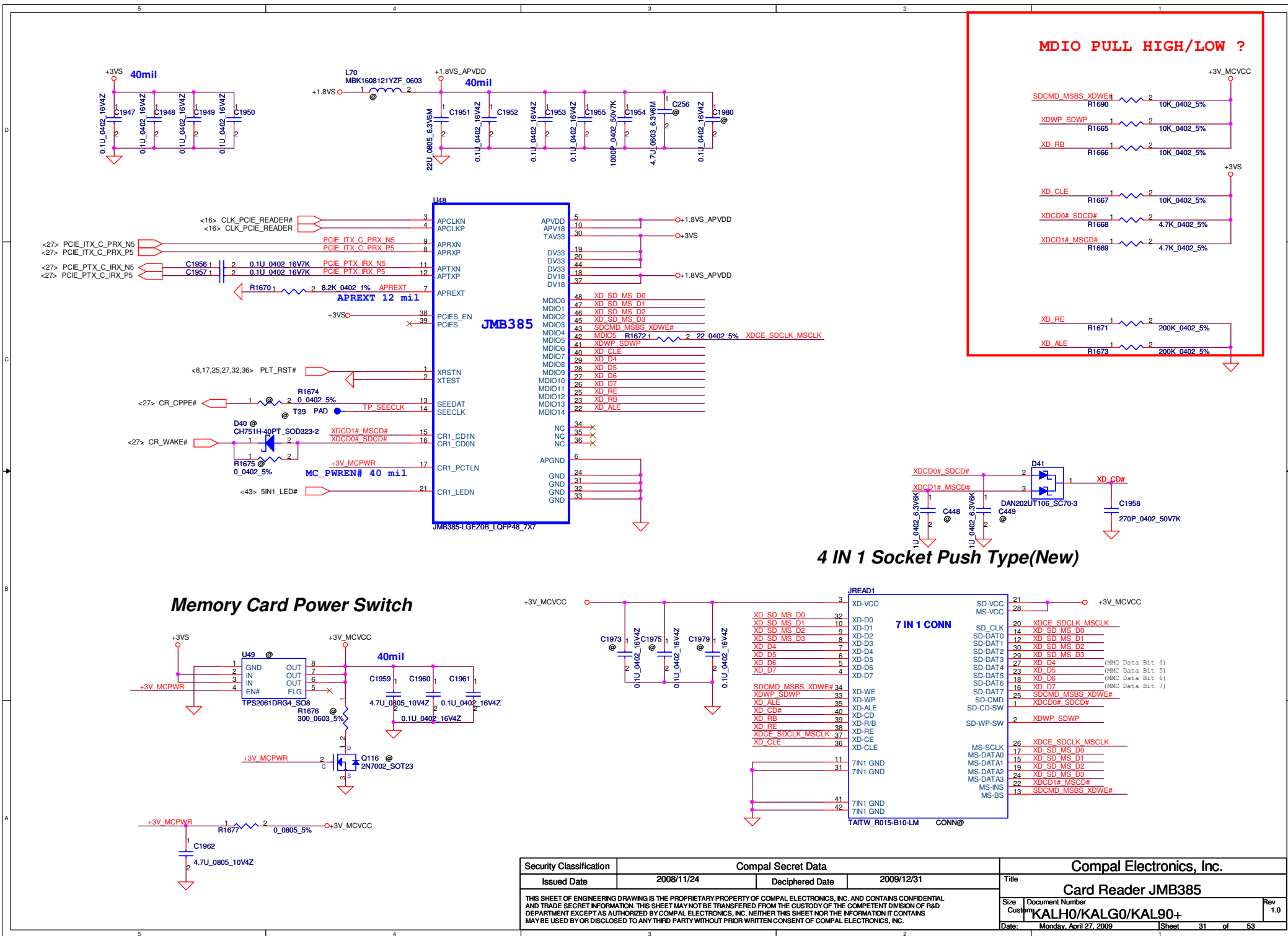
<26> SATA_DTX_C_IRX_N1
<26> SATA_DTX_C_IRX_P1

SATA_DTX_C_IRX_N1
SATA_DTX_C_IRX_P1

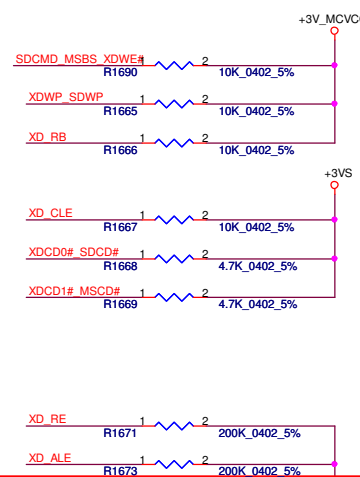
C1945
C1946

0.01U_0402_16V7K
0.01U_0402_16V7K

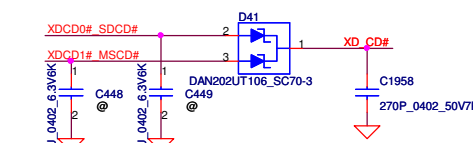
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Custom		KALH0/KALG0/KAL90+		1.0	
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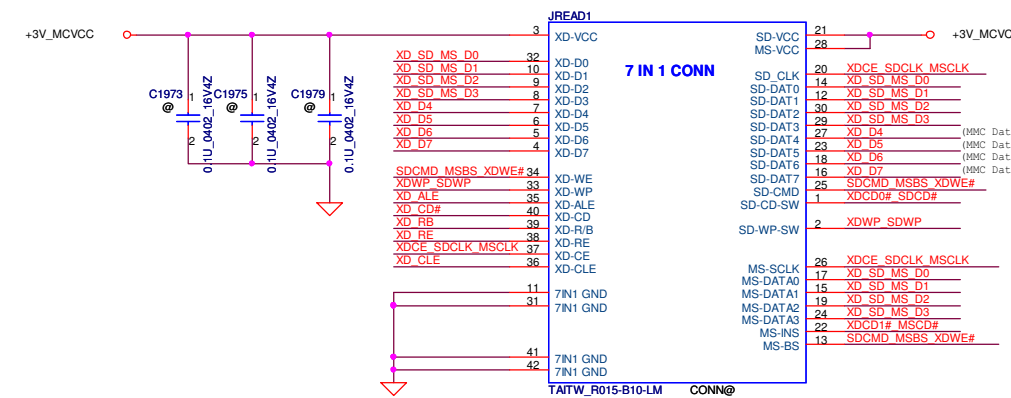
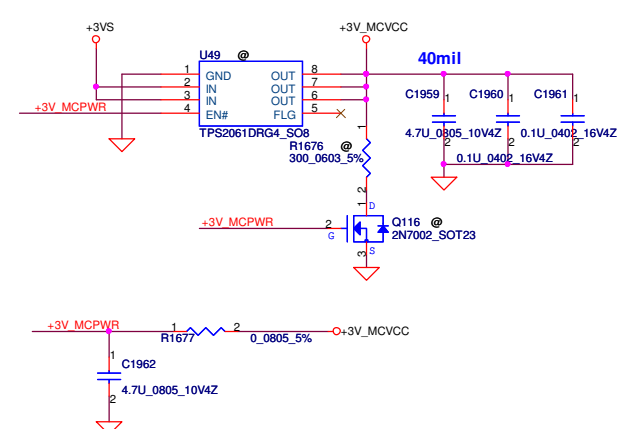
MDIO PULL HIGH/LOW ?



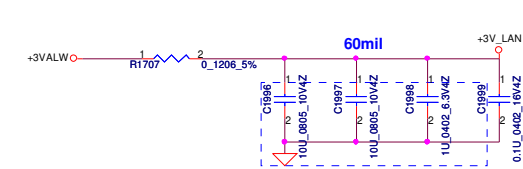
4 IN 1 Socket Push Type(New)



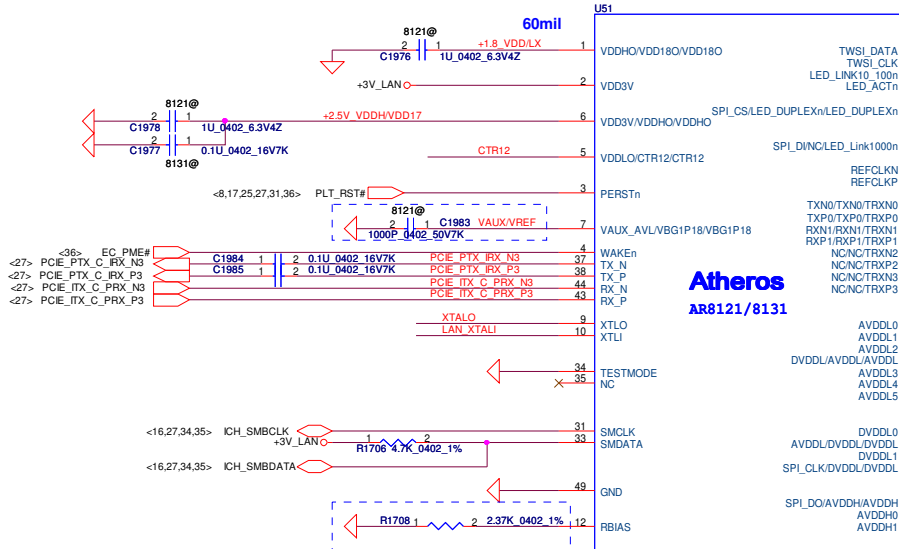
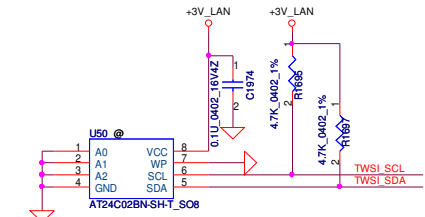
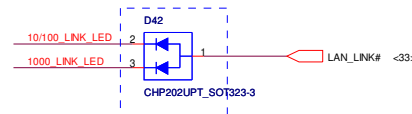
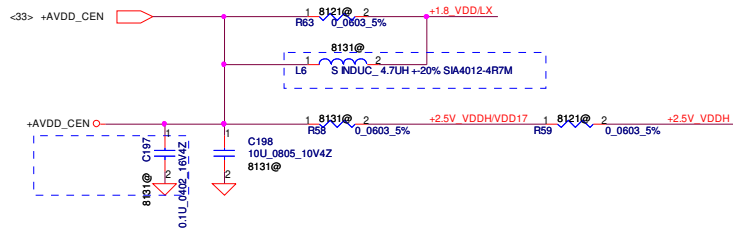
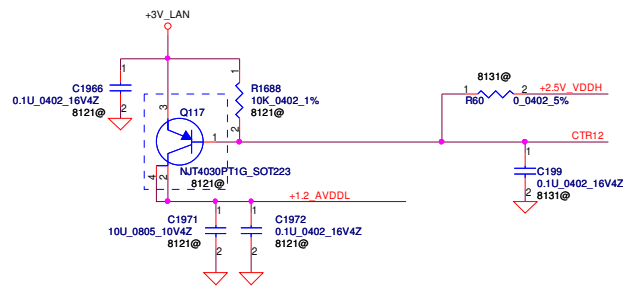
Memory Card Power Switch



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Customer	KALH0/KALG0/KAL90+	1.0		Monday, April 27, 2009	
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Place Close to Pin 2

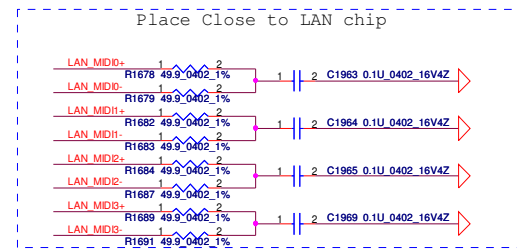
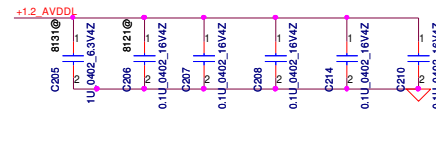
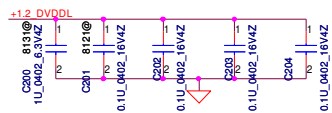
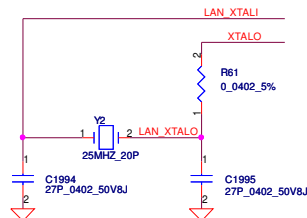


Atheros
AR8121/8131

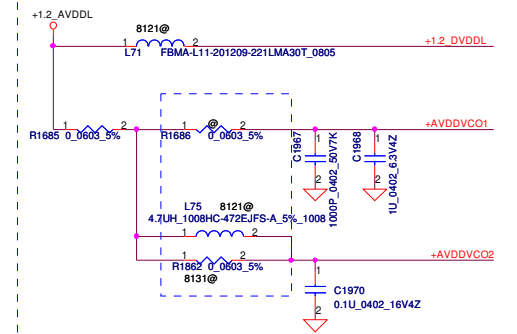
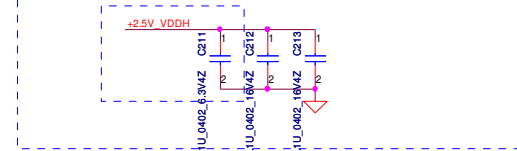
AR8121-AL1E QFN48 6X6
SA000031Z00 S IC AR8131-AL1E QFN 48P E-LAN CTRL
8121@
SA000025M00 S IC AR8121-AL1E QFN 48P E-LAN CTRL

Place Close to Pin 28、32、45、46
C1750 and C1730 close to Pin46
C1072 close to Pin45

Place Close to Pin8、16、22、36、39
C1066 and C1067 close to Pin8



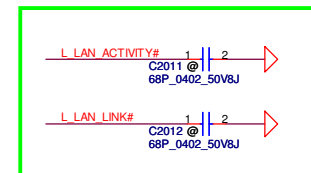
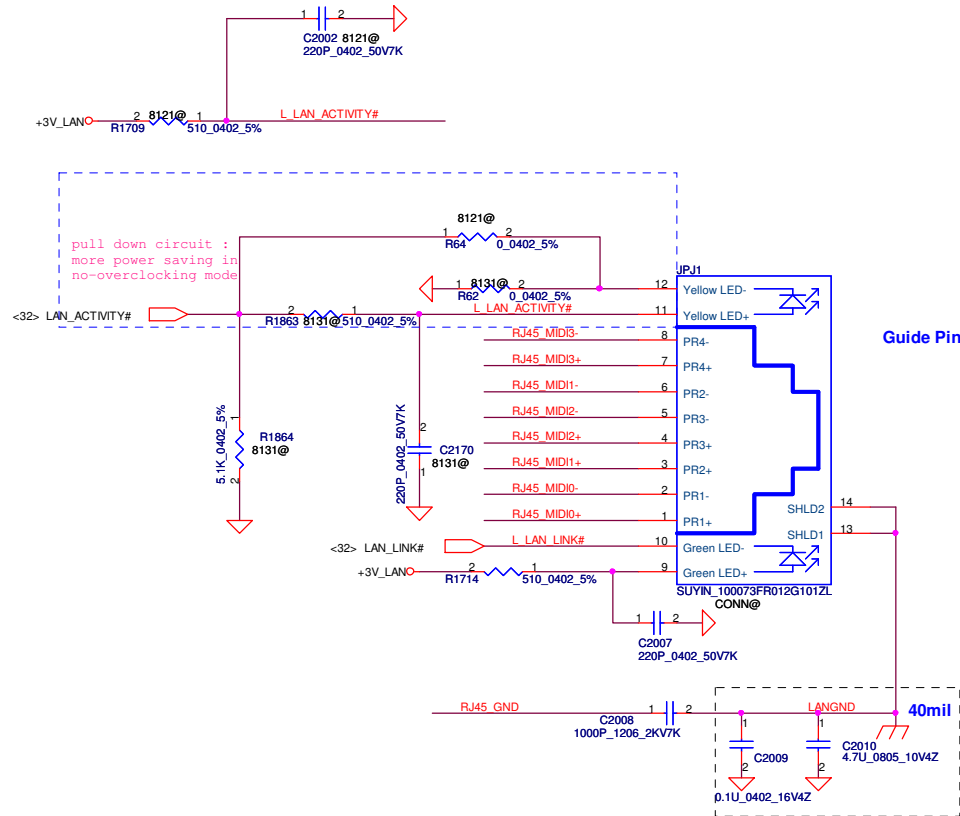
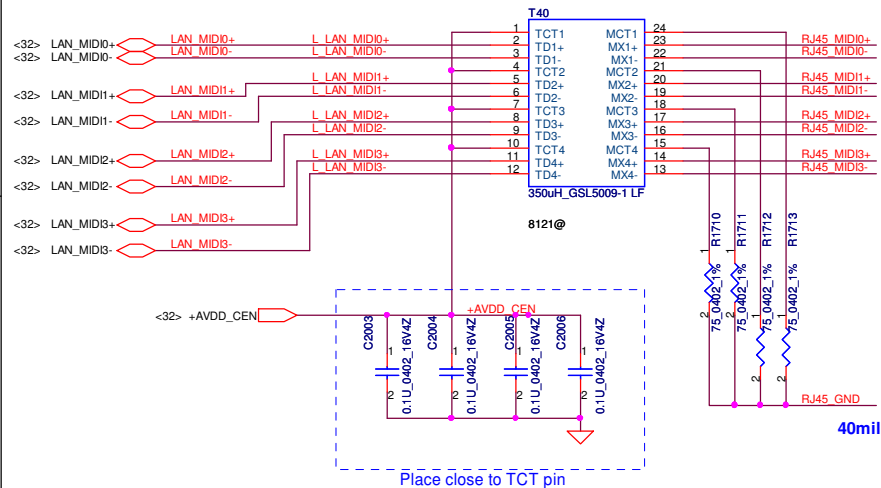
Place Close to Pin15、19、25
C1061 close to Pin15



8121 If not overclocking, R1685 & L75 stuffed and R1686 & R1687 removed
8131 If not overclocking, R1685 & R1682 stuffed and R1686 & L75 removed

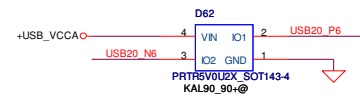
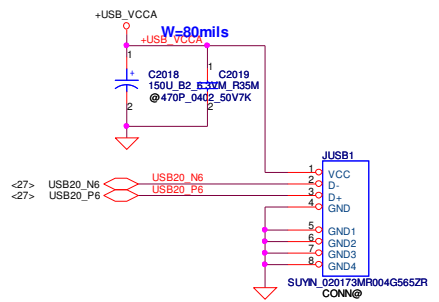
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Deciphered Date				2009/12/31				Atheros AR8131			
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LAN AR8121/8112



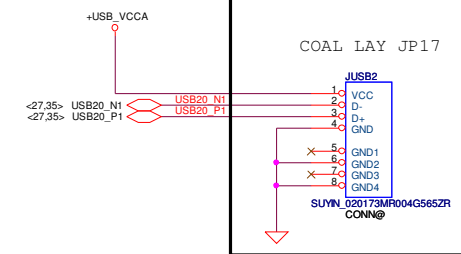
For EMI

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USB CONN.

The diagram illustrates the electrical connections for the FOX_AS0B226-S99N-7F module. It is divided into several functional sections:

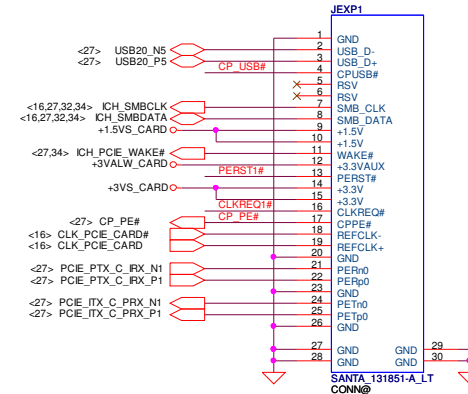
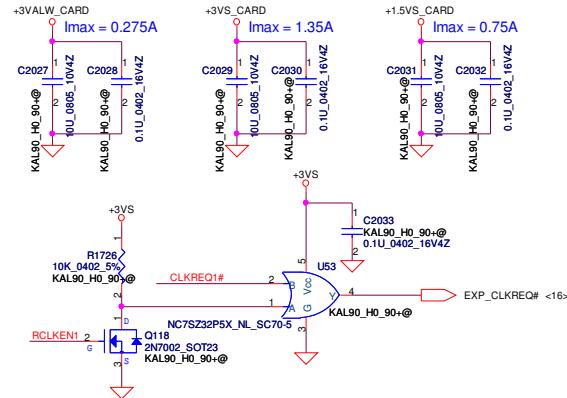
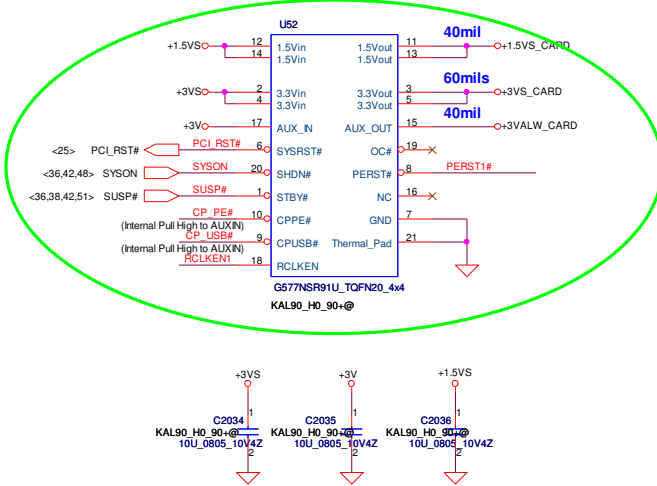
- Power and Ground Section:** Shows connections for +3VS_WLAN, +3V, +1.5VS, and +3V_WLAN. It includes capacitors C2020, C2021, C2022, C2023, C2024, and C2025, and resistors R1718, R1719, R1720, R1721, R1723, and R1724.
- Signal Section:** Lists various signal lines and their connections, including WLAN_BT_DATA, WLAN_BT_CLK, MIN2_CLKREQ#, CLK_PCE_MIN2, PCE_PTX_C_RX_N2, PCE_PTX_C_RX_P2, PCE_ITX_C_PRX_N2, PCE_ITX_C_PRX_P2, WL_OFF#, PLT_RST_BUF#, MIN2_SMBCLK, MIN2_SMBDATA, USB20_N11, USB20_P11, and MN1_LED#.
- Mini-Card Port Debug Section:** A dashed blue box highlights connections for E51TXD_P80DATA, E51RXD_P80CLK, and R1725.
- Module and Connector Section:** Shows the connection to the FOX_AS0B226-S99N-7F module and the CONN@ connector.



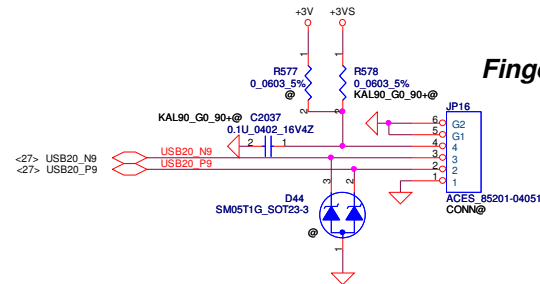
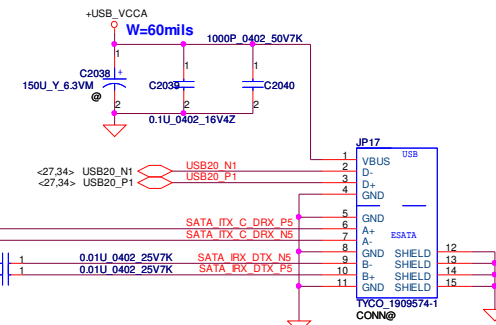
Pin 15 connection diagram for the ACES_85201-08051 component. The component is a blue rectangle labeled 'JP15' and '80mil'. It has 10 pins. Pin 1 is connected to a red wire labeled '+5VALW'. Pin 2 is connected to a red wire labeled 'SYSON# <35,42>'. Pin 3 is connected to a red wire labeled 'USB20_N0 <27>'. Pin 4 is connected to a red wire labeled 'USB20_P0 <27>'. Pin 5 is connected to a red wire labeled 'USB_OC#0 <27>'. Pin 6 is connected to a red wire labeled 'USB_OC#0 <27>'. Pin 7 is connected to a red wire labeled 'USB_OC#0 <27>'. Pin 8 is connected to a red wire labeled 'USB_OC#0 <27>'. Pin 9 is connected to a red wire labeled 'USB_OC#0 <27>'. Pin 10 is connected to a red wire labeled 'USB_OC#0 <27>'. The component is labeled 'ACES_85201-08051' and 'CONN#'. A red triangle symbol is at the bottom.

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				Size	Document Number		Rev
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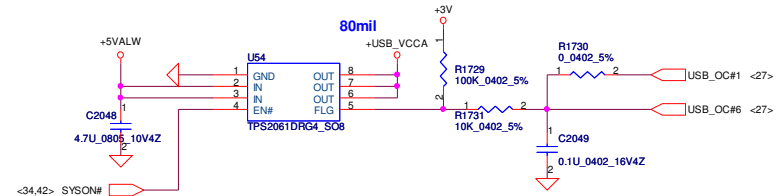
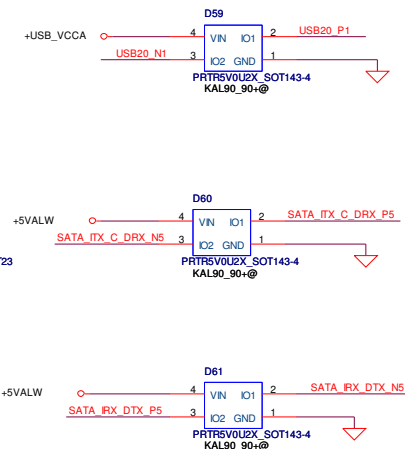
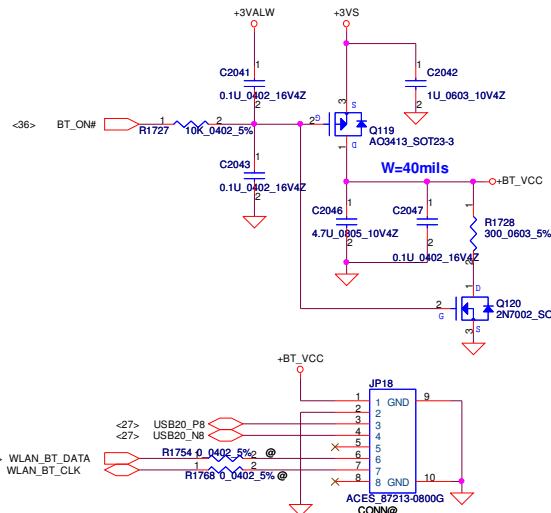
New Card Power Switch



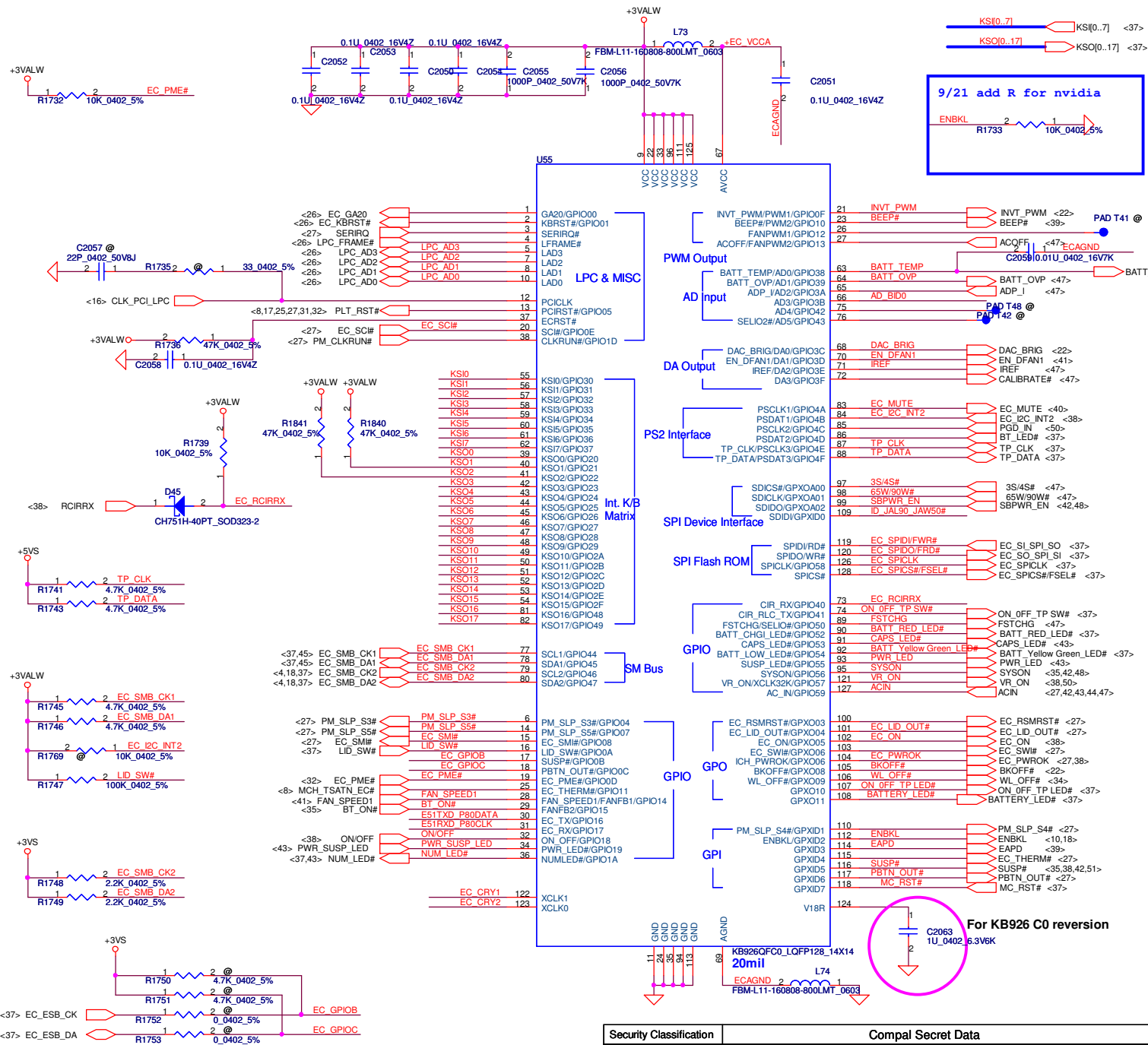
Finger Print Conn.

**ESATA CONN**

Bluetooth Conn.

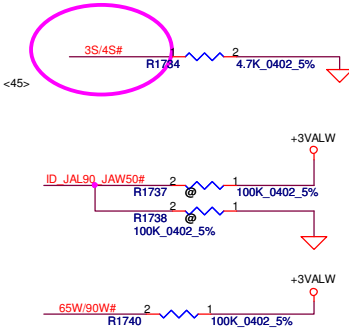


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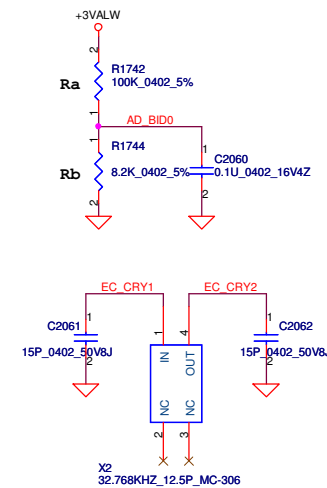


For EC Tools

Place on RAM door

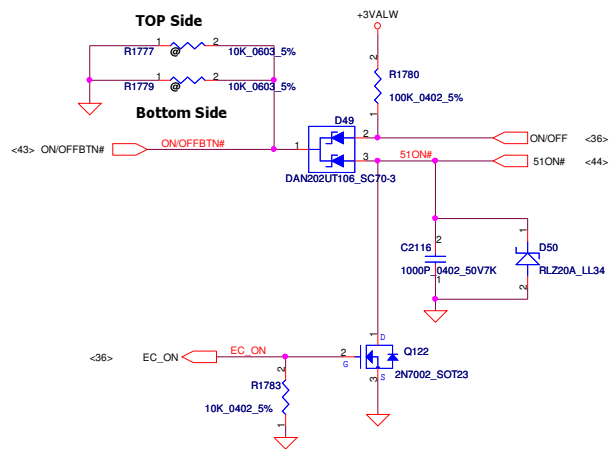


Analog Board ID definition, Please see page 3.



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ON/OFF switch

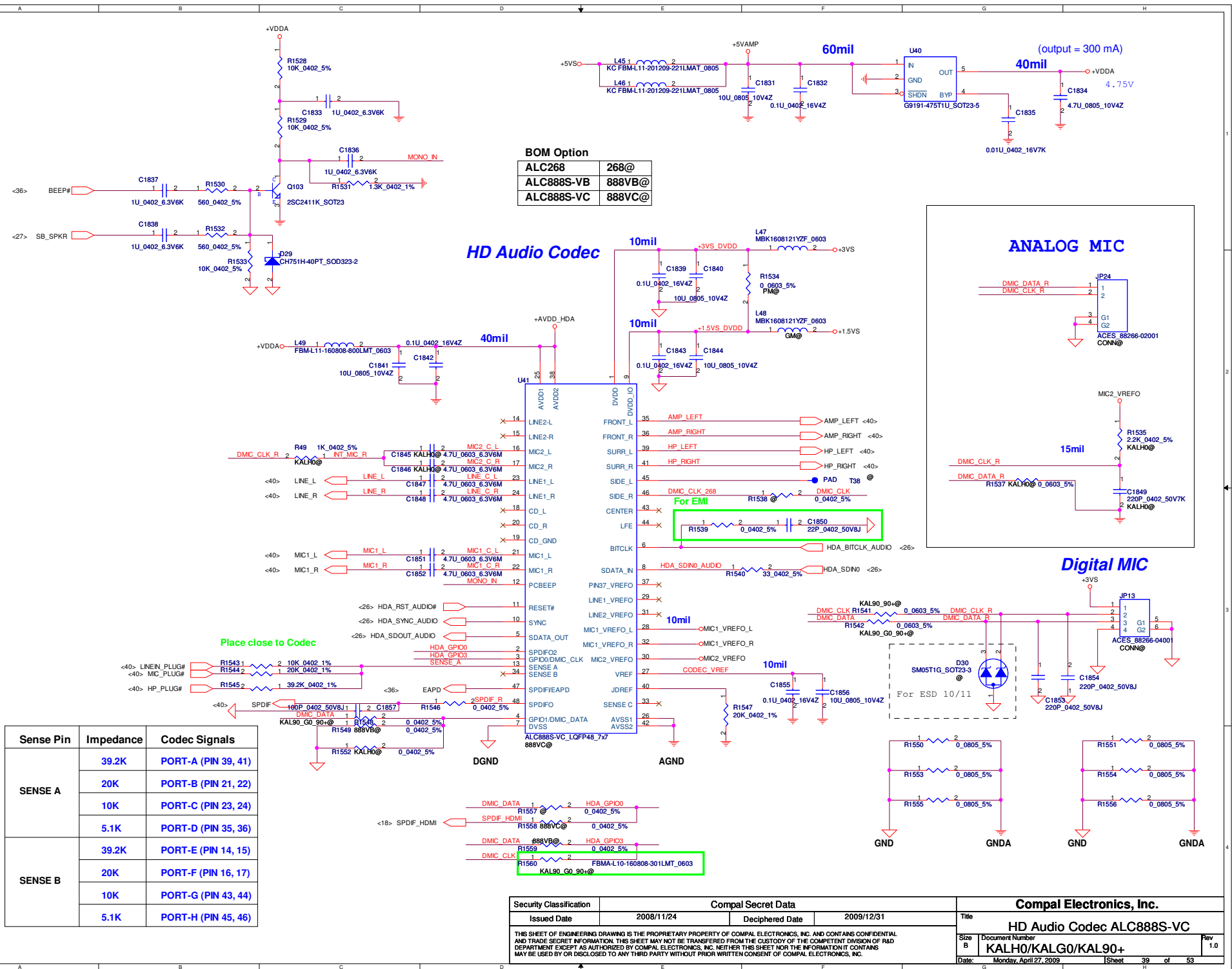


For South Bridge

The schematic diagram illustrates the power management section for the South Bridge. It features two voltage regulators, U59A and U59B, which are SN74LVC14APWLE_TSSOP14. These regulators are powered by +3VALW and +3VS. The output of U59A is connected to the input of U59B, which then drives the EC_PWROK signal. The output of U59B is connected to the input of U59C, which then drives the VS_ON signal. The output of U59C is connected to the input of U59D, which then drives the VS_ON signal. The diagram also shows various resistors (R1784, R1785, R1787, R1786) and capacitors (C2117, C2118, C2119) used for signal conditioning and timing. The input signals are VR_ON, SUSP#, and SUSP, and the output signals are EC_PWROK and VS_ON.

For +VCCP/+1.05VS

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				Date:	Monday, April 27, 2009
				Sheet	38 of 53

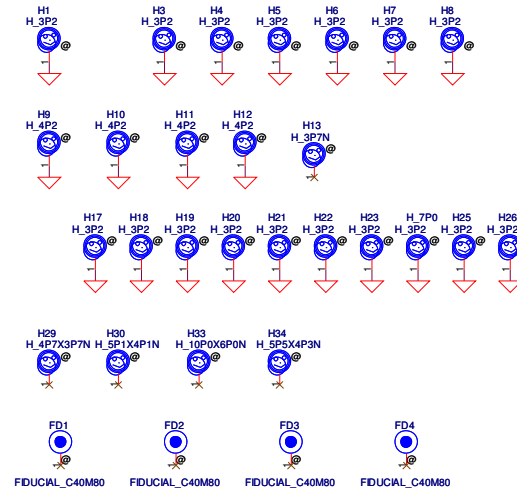
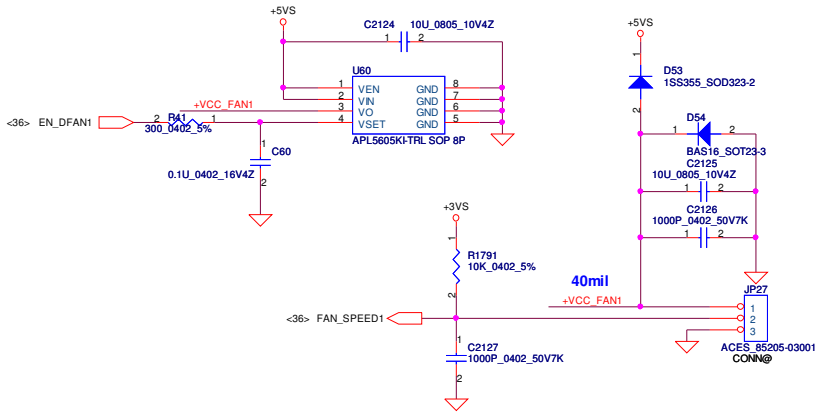


BOM Option	
ALC268	268@
ALC888S-VB	888VB@
ALC888S-VC	888VC@

Sense Pin	Impedance	Codec Signals
SENSE A	39.2K	PORT-A (PIN 39, 41)
	20K	PORT-B (PIN 21, 22)
	10K	PORT-C (PIN 23, 24)
	5.1K	PORT-D (PIN 35, 36)
SENSE B	39.2K	PORT-E (PIN 14, 15)
	20K	PORT-F (PIN 16, 17)
	10K	PORT-G (PIN 43, 44)
	5.1K	PORT-H (PIN 45, 46)

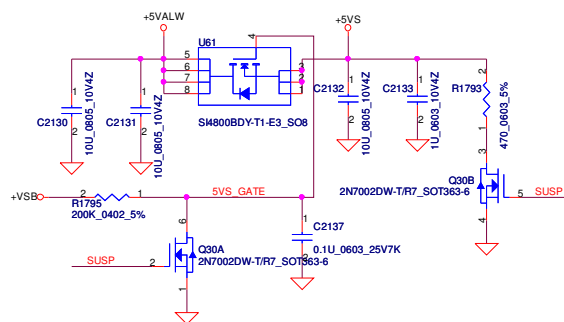
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Issued Date	2008/11/24	Deciphered Date	2009/12/31	Title	
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Size B	Document Number	KALHO/KALGO/KAL90+		Date	Monday, April 27, 2009
		Sheet	39	of 53	

FAN1 Conn

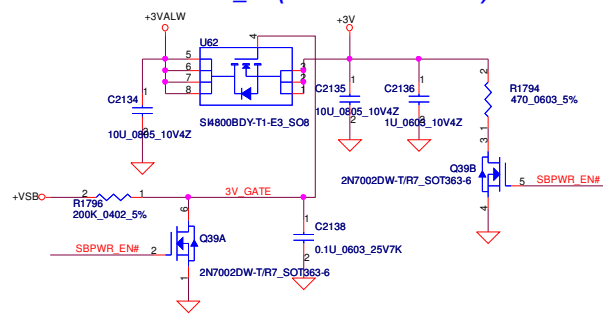


Security Classification		Compal Secret Data		Compal Electronics, Inc.	
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Size		Document Number			Rev
B		KALH0/KALG0/KAL90+			1.0
Date:		Monday, April 27, 2009		Sheet 41 of 53	

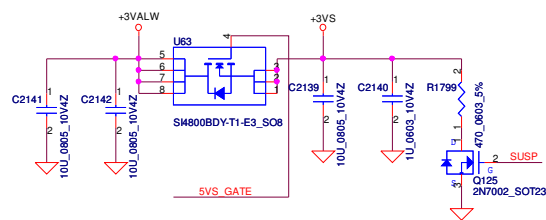
+5VALW TO +5VS



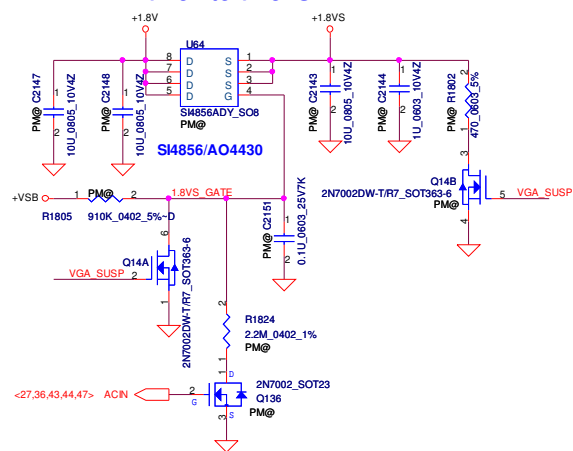
+3VALW TO +3V_SB(ICH8M AUX Power)



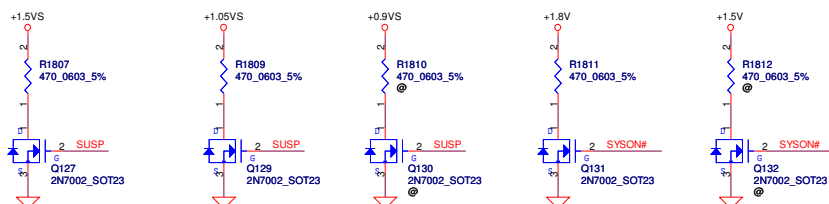
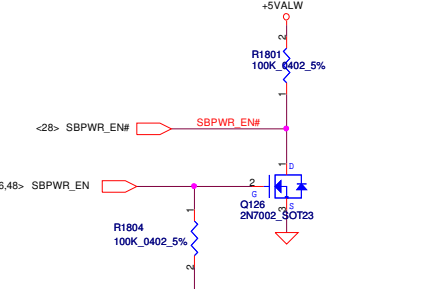
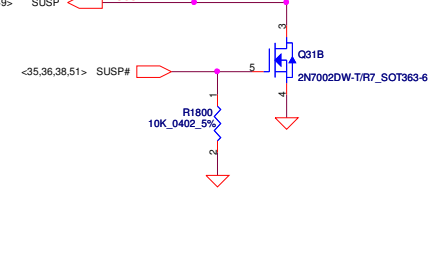
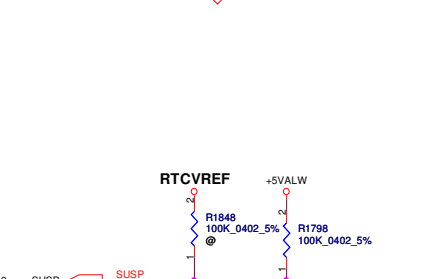
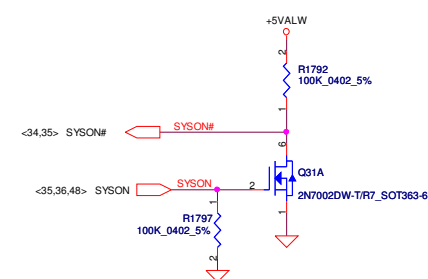
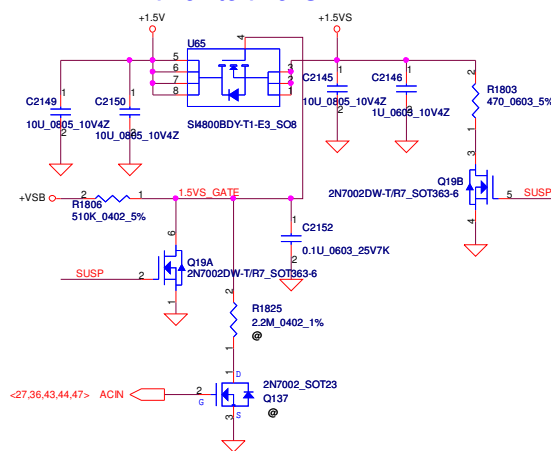
+3VALW TO +3VS



+1.8V to +1.8VS

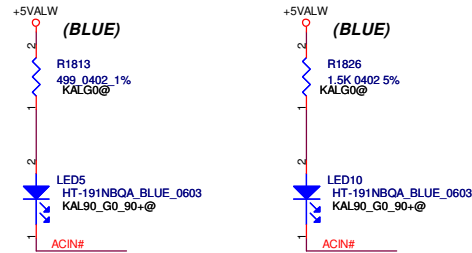


+1.5V to +1.5VS

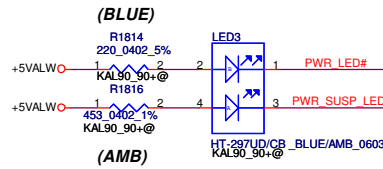


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Size B	Document Number	KALHO/KALGO/KAL90+		Date	Monday, April 27, 2009
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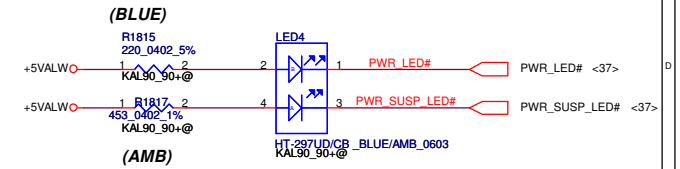
Enlightener LED



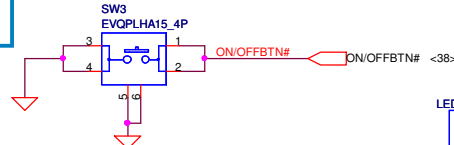
ON/OFF LED LEFT



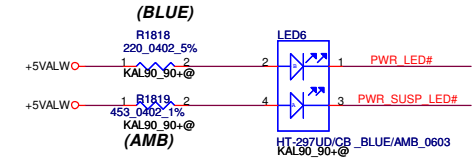
ON/OFF LED RIGHT



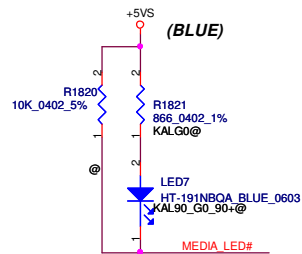
ON/OFF Button



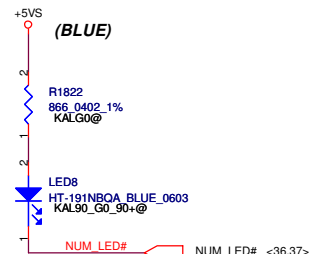
ON/OFF LED DOWN



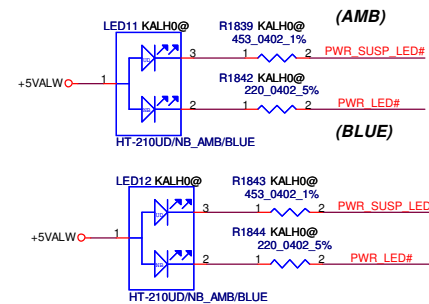
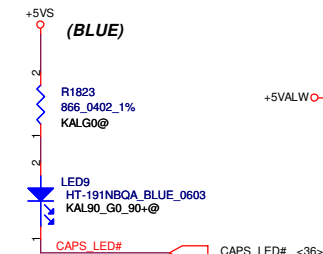
MEDIA_LED



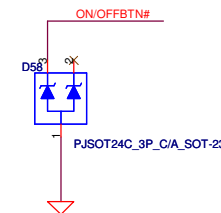
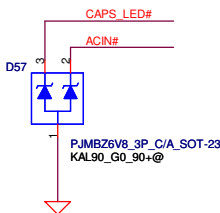
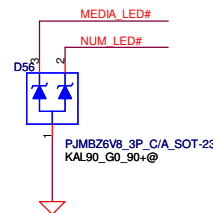
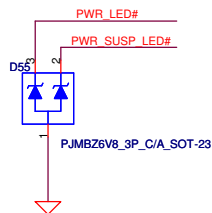
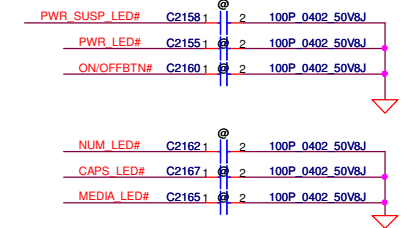
NUM_LED



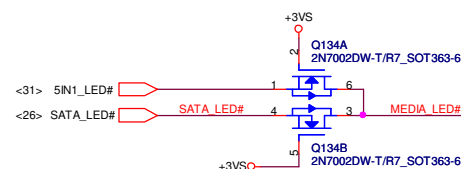
CAPS_LED



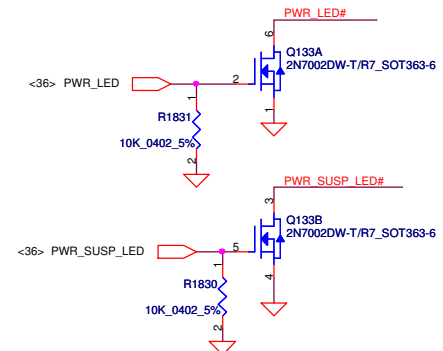
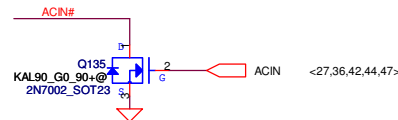
FOR EMI



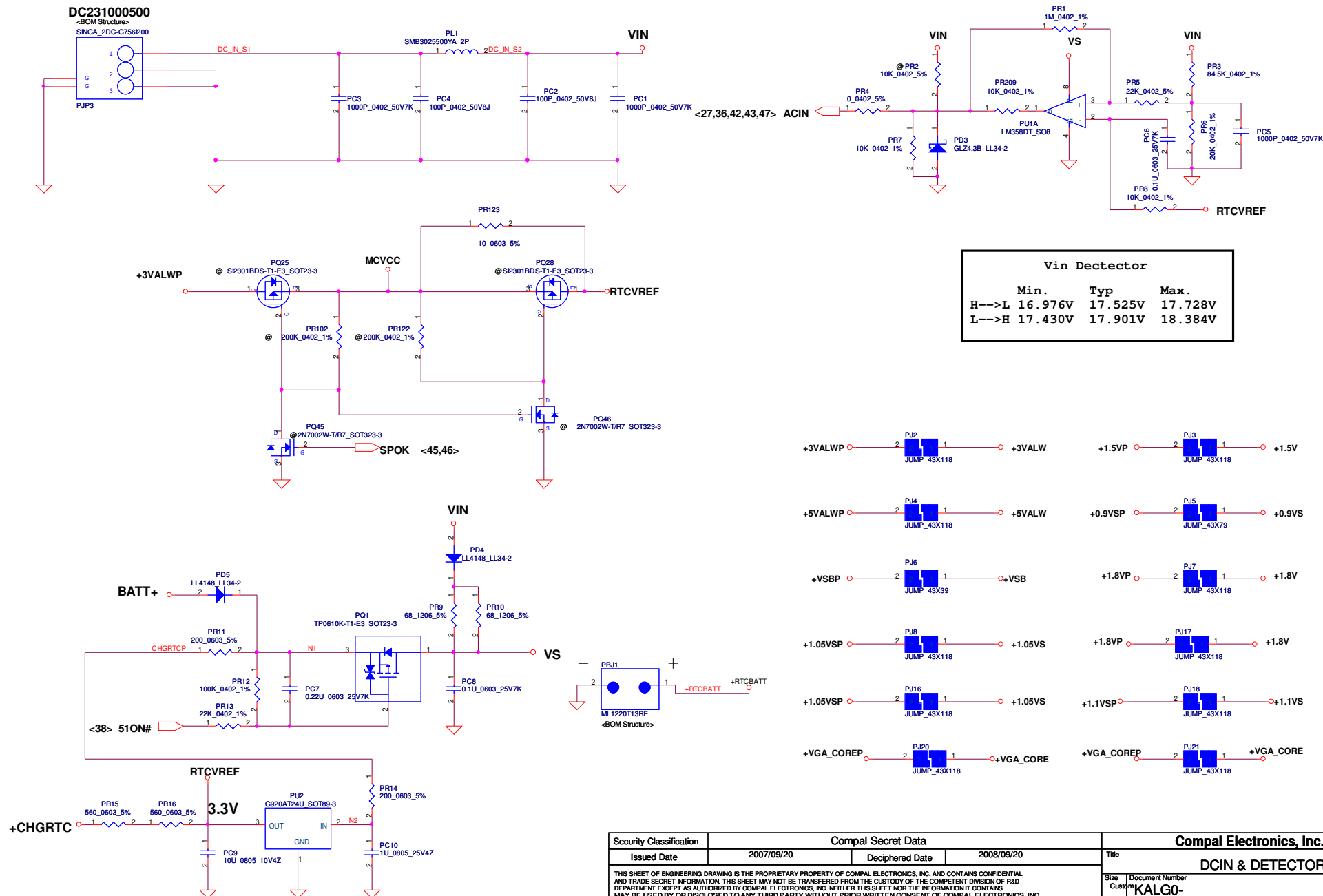
D1 D2 D3 USE PANJIT PJMBZ6V8
SCA00000100
6.8V



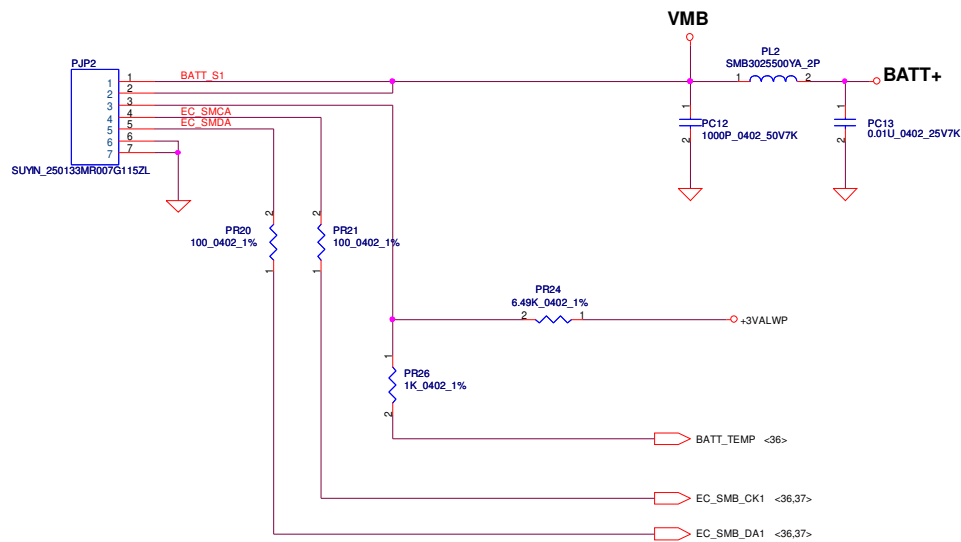
D4 USE
PJSOT24C 3P C/A SOT-23
SCA00000E00
24V



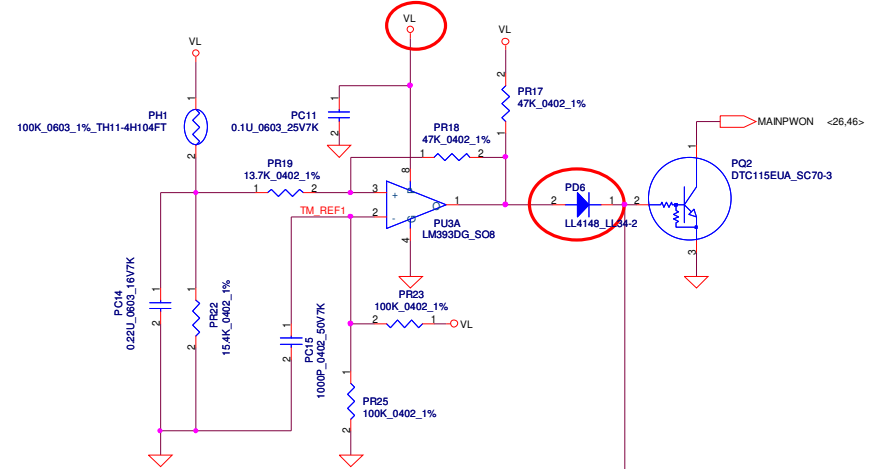
Security Classification		Compal Secret Data		Compal Electronics, Inc.	
Issued Date	2008/11/24	Deciphered Date	2009/12/31	Title	PWR/B
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				Document Number	KALH0/KALG0/KAL90+
				Date	Monday, April 27, 2009
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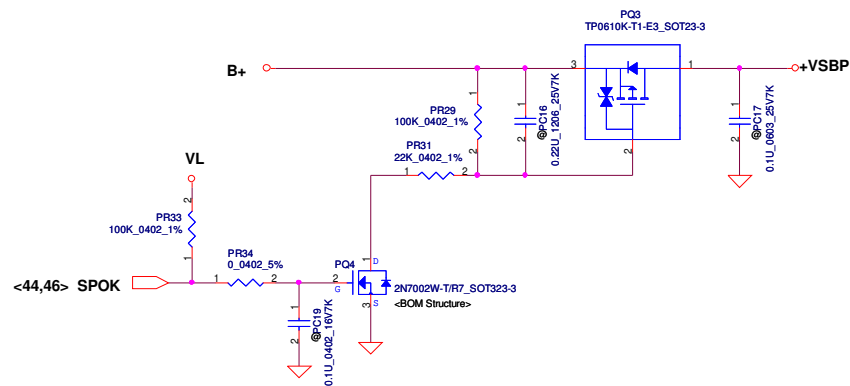
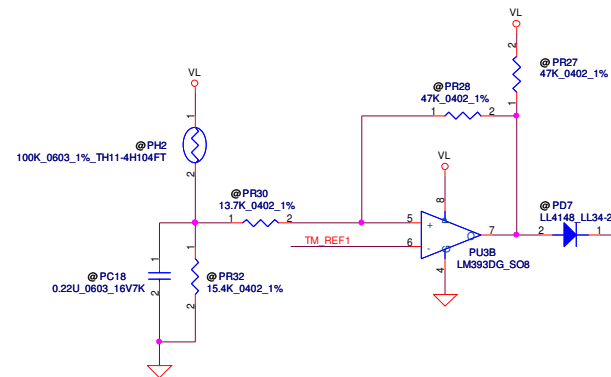
Security Classification	Compal Secret Data		Compal Electronics, Inc.	
Issued Date	2007/09/20	Deciphered Date	2008/09/20	Title
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Size	Document Number	Rev		
Custom	KALG0-	0.1		
Date:	Monday, April 27, 2009	Sheet	44	of 53



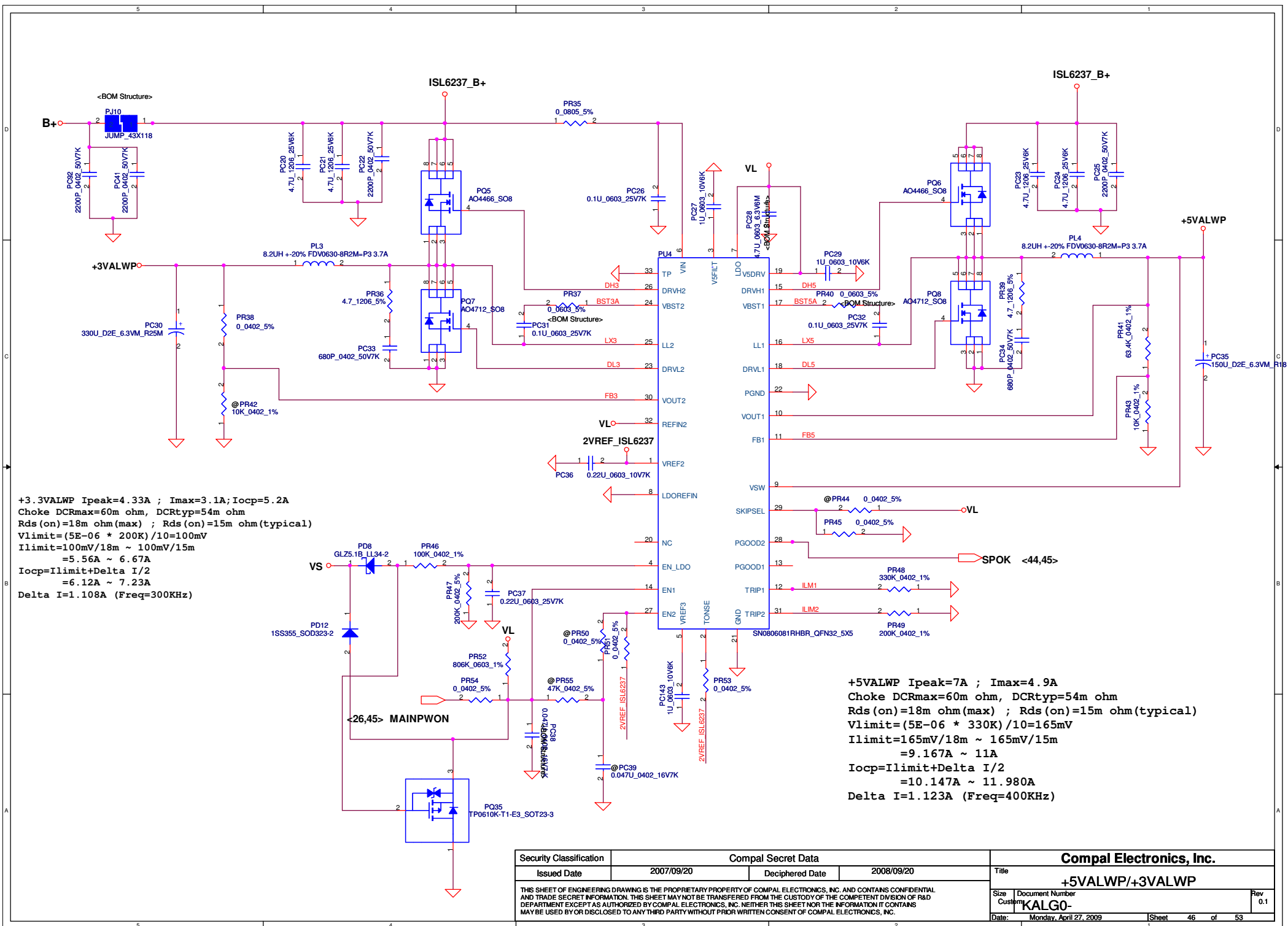
PH1 under CPU botten side :
 CPU thermal protection at 96 degree C
 Recovery at 60 degree C



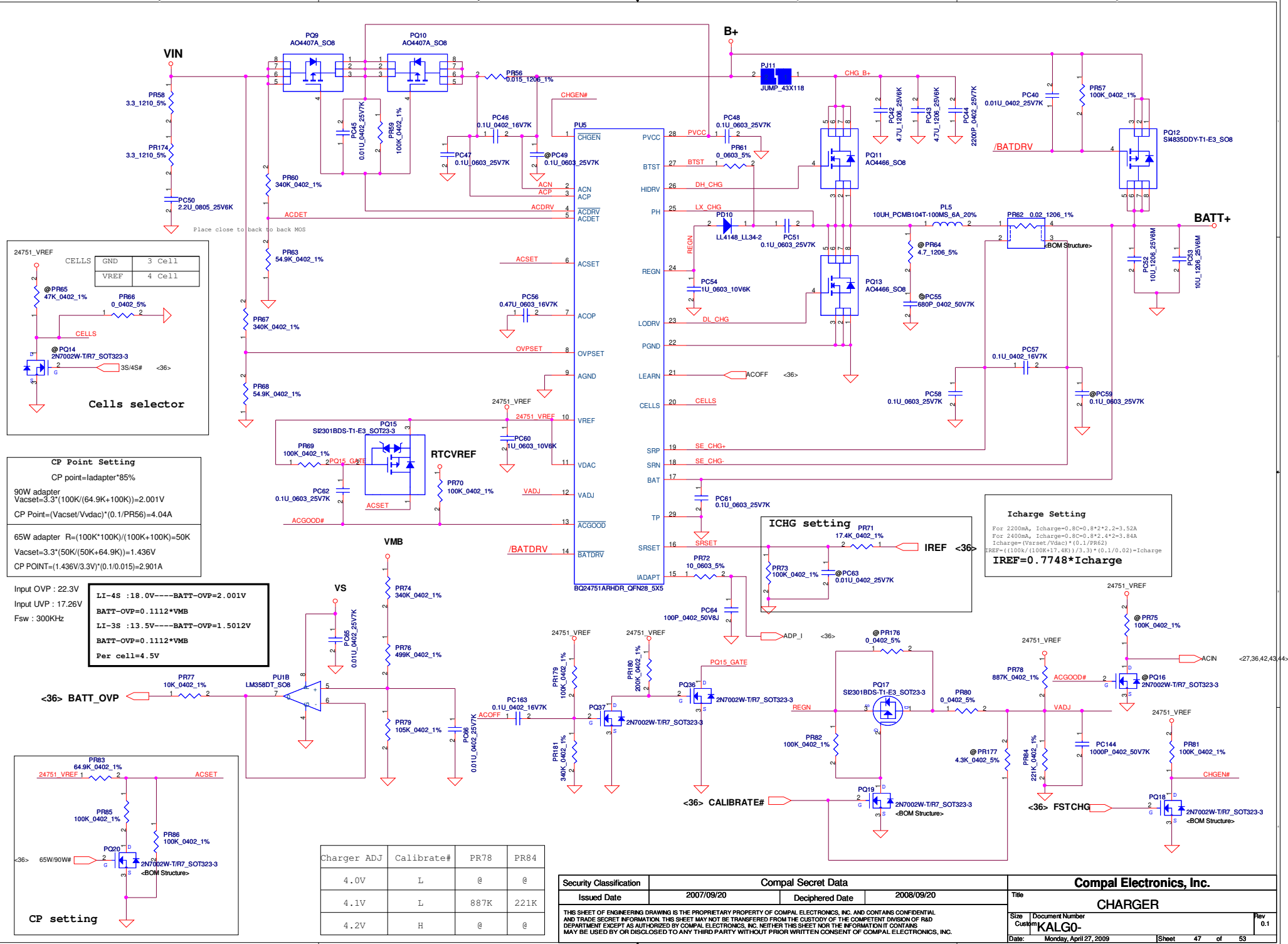
PH2 near main Battery CONN :
 BAT. thermal protection at 79 degree C
 Recovery at 47 degree C

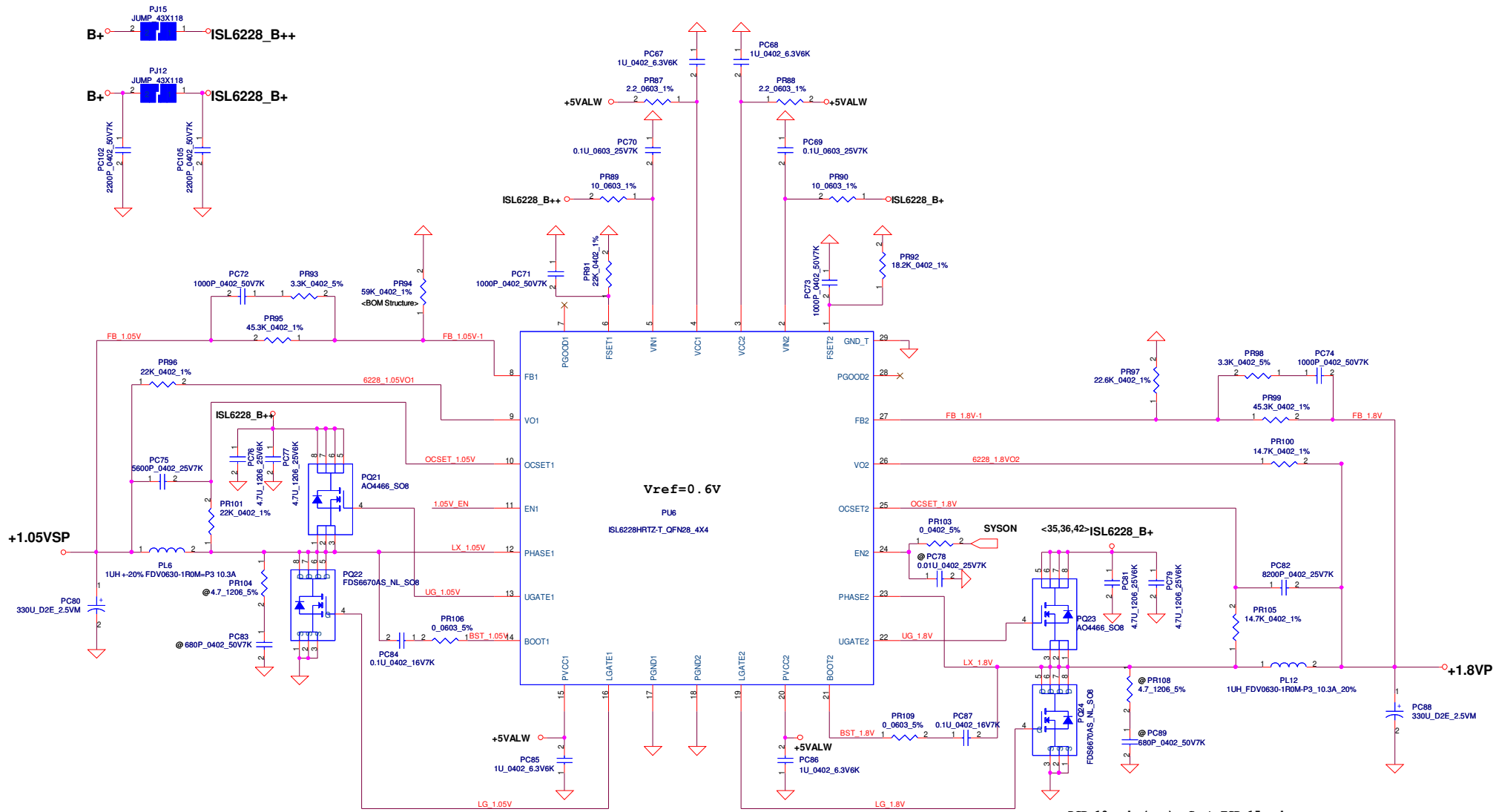


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Size	Custom	Document Number	KALG0-	Rev	0.1
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Issued Date	2007/09/20	Deciphered Date	2008/09/20	Title	+5VALWP/+3VALWP	
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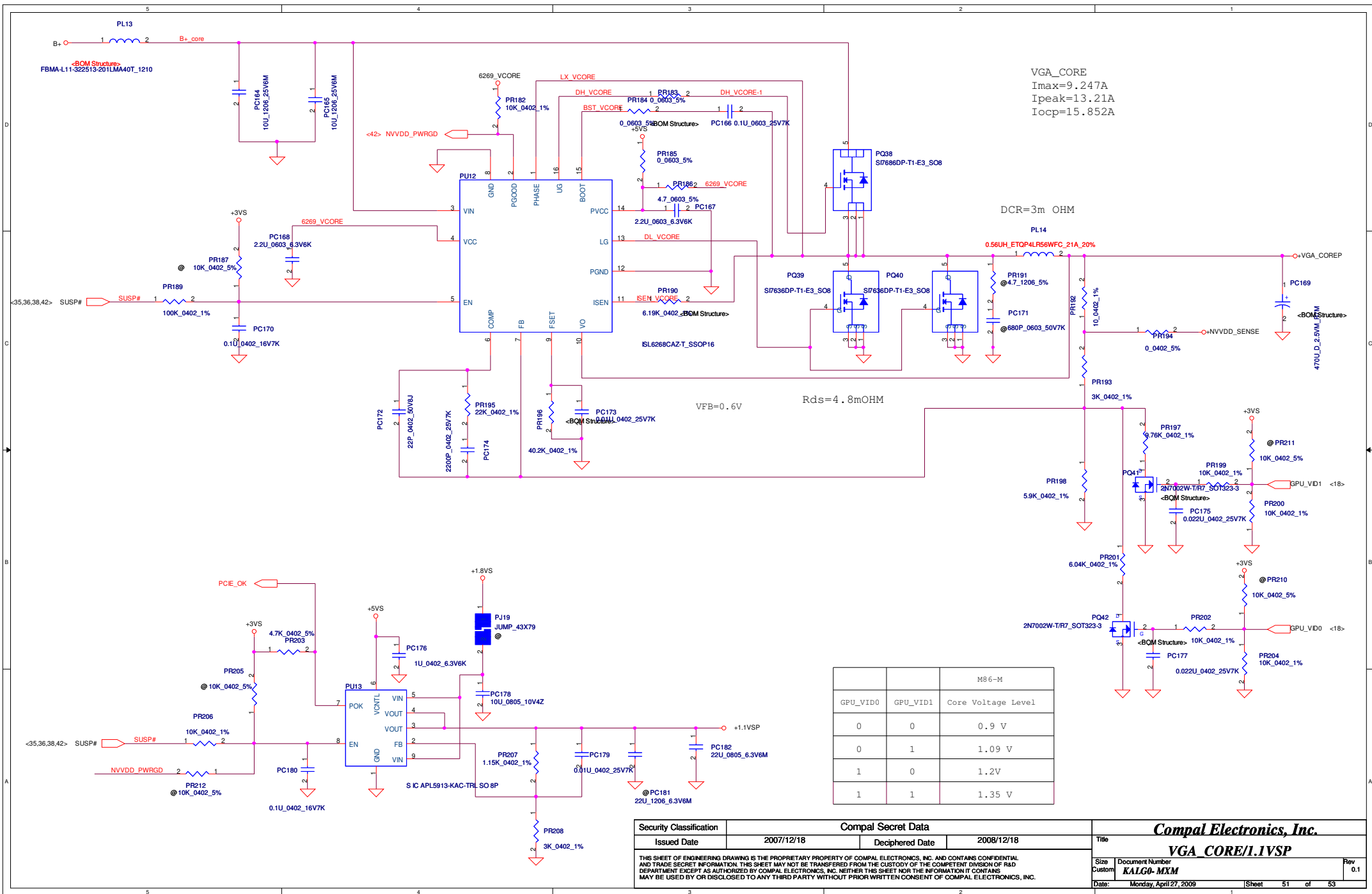
DCR 10m ohm(max) Cout ESR=15m ohm
 1.05VSP (Ibudget=17.47A) OCP Setting
 $F_{sw}=1/1.5E-10*22k=303K$
 $V_o=V_{ref} * ((PR95+PR94)/PR94)$
 $I_{peak}=13.29A$, $I_{max}=9.31A$
 $I_{ocp}=13.29*1.2=15.95A$
 $\Delta I=4.09A$
 $I_{ocp}*DCR=(Rocset*9.5uA)=(15.95*1.3*10m; Roset=21.8K$
 now chose Roset=22K
 $C_{sen}=L/(DCR*Roset)=1uH/(10m*22k)$; $C_{sen}=0.00523uF$
 now chose $C_{sen}=5600pF$
 $I_{ocp_min}=(22K*9.5uA)/(10m\ ohm*1.3)=16.07A$
 $I_{ocp_max}=(22K*10.5uA)/(10m\ ohm)=23.1A$

<36,42> SBPWR_EN

<38,49> VS_ON

DCR 10m ohm(max) Cout ESR=15m ohm
 1.8V (Ibudget=9.91A) Ipeak=9A, Imax=6.3 A
 $F_{sw}=1/1.5E-10*18.2k=366K$
 $V_o=V_{ref} * ((PR97+PR99)/PR97)$
 $I_{ocp}=9*1.2=10.8A$
 $I_{ocp}*DCR=(Rocset*9.5uA)=10.67*1.3*10m; Roset=14.78K$
 now chose Roset=14.7K
 $C_{sen}=L/(DCR*Roset)=1uH/(10m*14.7k)$; $C_{sen}=8.16nF$
 now chose $C_{sen}=8200pF$
 $I_{ocp_min}=(14.7K*9.5uA)/(10m\ ohm*1.3)=10.74A$
 $I_{ocp_max}=(14.7*10.5uA)/(10m\ ohm)=15.43A$

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VGA_CORE
Imax=9.247A
Ipeak=13.21A
Iocp=15.852A

DCR=3m OHM

VFB=0.6V

Rds=4.8mOHM

		M86-M
GPU_VID0	GPU_VID1	Core Voltage Level
0	0	0.9 V
0	1	1.09 V
1	0	1.2V
1	1	1.35 V

Security Classification		Compal Secret Data		Compal Electronics, Inc.											
Issued Date		2007/12/18		Deciphered Date		2008/12/18		Title							
								VGA_CORE/1.IVSP							
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								Custom	KALGO-MXM	0.1					
								Date:	Monday, April 27, 2009			Sheet	51	of	53

Version change list (P.I.R. List)

Page 1 of 2
for PWR

Item	Fixed Issue	Reason for change	Rev.	PG#	Modify List	Date	Phase
1	Change PR196 value	VGA OCP	0.3	50	change the resistance value of pr196 from 57.6K to 40.2K	2009/02/06	PVT
2	Change PL14 value	prevent OVP occur	0.3	50	change the inductance value of pl14 from 1uH to 0.56 uH	2009/02/06	PVT
3	Change PU4 IC part number	vender suggestion	0.3	45	change part number to SA00002V400	2009/02/06	PVT
4							
5							
6							

Security Classification		Compal Secret Data		Compal Electronics, Inc.	
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				PIR (PWR)	
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B --> C Change List

0220-----

Page 6, Change C98 BOM structure to @

0218-----

Page 20, Add R569 with BOM structure @

Page 23, R34, R38, R40, R53, R54, R55 change BOM structure to @

Page 31, Change R215 to 18K

Update Power Schematics

0213-----

Page 11, Delete R113

Add J1 JUMP_43X79 with BOM structure @

Page 31, R558, R559 0 ohm with BOM structure @

R560, R561 4.7K with BOM structure @

Page 32, R562, R563 0 ohm with BOM Structure @

R564, R565 0 ohm

Page 38, Delete C619

Add R566, R567, R568 0_0603 with BOM Structure JAL90@

0204-----

Page 12, Change L31 to MBK1608301YZF_0603 with BOM structure GM@

Change R163 to 0_0805_5%

Page 23, Change BOM Structure of U5 to @

Page 27, Change BOM Structure of R555 and R550 to @

Page 33, Change R217 to 31.6K_0402_1%

Change C313 to 1U_0603_10V6K

Page 34, Change R503 to FBMA-L10-160808-301LMT_0603

01/31-----

Page 23, Delete U10

01/29-----

Page 23, Change R152 BOM Structure to @

01/24-----

Page 4, Change U8 to SA00001Z700 (EMC1402)

Page 33, Change C338 to SE076104K80

Page 35, Mount C584

01/23-----

Page 38 Delete F3, R558~560, C609~614

01/22-----

Page 11, Delete R79

Change J1 Symbol to JUMP_43X79

Page 33, Add R557 10K (Check)

Change R245 BOM Structure with @

Page 38, Add C609~614, R558~560 (Check)

C607,608, 615~619 (Check)

01/17-----

Page 11, Add R79 0_0805

Update Power Schematics

01/16-----

Page 11, Delete R79 0_0805

Add J1 JUMP_43X79

Page 16, Change C296, R301 to 27P_0402

Page 19, Change L17, L19, L21 BOM structure to GM@

Page 23, Mount U29, R339

Add U10 with BOM structure @ (Co-lay with U5)

Change R340 Bom structure to @

Change U5 to MX25L4005AMC-12G_SO8 (SA00002A900)

Page 27, Change U26, C420 BOM structure to @

Change R550 to 0_0402

Add R555 0_0402

Page 32, Change R269 to 240_0402_5%, R267 to 453_0402_1%

Change R268 pin1 connect to +5VALW

Page 33, Change R217 to 18K_0402_1% with BOM structure PM@

Page 35, Add R551,R552, R553, R554 75_0603_1% with BOM structure JAL90@

Add D32 PJDLC05_SOT23-3

Page 38, Add F3 3A_15VDC_SMD2920P300TF/15

Page 49, Add R551,R552, R553, R554 1K_0603_1% with BOM structure 268@

Add L17, L19, L21 0_0805 with BOM structure PM@

Update U38 (ALC268-VB1-GR) PN:SA00001GD10 for JAW50

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